

# DeXposure-Claw: An Agentic System for DeFi Risk Monitoring and Decision Recommendation Based on a Graph Time-Series Foundation Model

Aijie Shu and Fengxiang He

**Abstract**—Decentralised finance (DeFi) has become a densely interconnected financial system in which credit exposures propagate through tokenised liabilities, making timely *risk monitoring* and *decision recommendation* essential for investors, protocol operators, and regulators. Existing agentic pipelines for DeFi risk rely largely on static heuristics or general-purpose LLM reasoning, and lack a forecasting backbone for graph-structured exposure dynamics. We propose a forecast-grounded regulatory decision framework: forecast future exposure graphs with DeXposure-FM, measure network risk and scenario stress deterministically, translate the evidence into supervisory tickets, and gate intervention actions by data health and confidence. To evaluate this framework class, we introduce *DeXposure-Bench*, a six-axis benchmark for FM + LLM decision pipelines on temporal exposure graphs, with regulator-aligned absolute-loss ground truth and eight reference method implementations. We instantiate the framework as *DeXposure-Claw*, a deployable agentic system with ticket-level audit logs. On the frozen 2025 decision split, DeXposure-Claw raises ticket F1 from 0.0076 (persistence + rules) to 0.0233 (+207%) at a measurable false-intervention cost (FIR  $\approx$  0.44) compared with rule-based baselines (FIR = 0). A tier-above Claude Opus 4.8 judge—with a cross-family panel of Gemini 2.5 Pro and GPT-5.5—confirms the headline ordering across judge families. Swapping the decision LLM for a tier-below Claude Sonnet 4.6 *strictly dominates* the Opus 4.7 baseline on every LLM-bearing axis: F1 rises further to 0.0288, FIR collapses to 0.000, judge quality reaches 2.65/5, at  $\sim 5\times$  lower inference cost. The benchmark therefore characterises DeXposure-Claw as a deployable Pareto option under the headline Opus 4.7 configuration, with a strict improvement available under the cheaper Sonnet 4.6 decision LLM. We release the harness, dataset, reference implementations, and ticket-level audit logs to support extension.

**Index Terms**—Agentic system, graph, time series, foundation model, decentralised finance, financial network, risk modelling.

## 1 INTRODUCTION

In May 2022, the Terra/Luna collapse erased over \$40 billion in value within 48 hours, triggering cascading liquidations across lending protocols, DEXs, and yield aggregators that held UST-denominated assets. Five months later, the FTX insolvency exposed hidden cross-protocol exposures and wiped out another \$8 billion. These crises share a common pattern: shocks propagated through an interconnected web of tokenised liabilities—credit exposures—that no single participant fully observed. In decentralised finance, exposures are high-dimensional, time-varying, and often implicit; data are noisy and incomplete; and crises unfold within days, leaving limited time for manual assessment.

Automated *agentic* pipelines offer a principled response: an agent can ingest on-chain measurements, maintain a belief over the evolving exposure network, surface early-warning signals, stress-test plausible shocks, and recommend risk-mitigating actions with auditable rationale. However, existing agentic proposals for DeFi risk largely rely on general-purpose large language models or static heuristics, and lack a dedicated forecasting backbone for graph-structured time series. Without accurate and data-efficient

prediction of network dynamics, monitoring agents suffer from high false-alarm rates and delayed detection, while decision agents risk recommending actions based on brittle or poorly calibrated beliefs. In high-stakes contexts, uncertainty calibration and robustness to missingness and regime shifts are not optional—they determine whether an agent is trusted and deployable.

This paper addresses that bottleneck by formulating a forecast-grounded regulatory decision framework for DeFi exposure networks. We leverage DEXPOSURE-FM, a foundation model trained on over 43.7 million data entries from DeFi exposure networks, to forecast the evolution of credit-exposure graphs across multiple horizons. The framework then follows a *forecast-measure-decide-gate* paradigm: predicted graphs are converted into risk measures and scenario evidence, decision policies turn this evidence into supervisory tickets, and data-health/confidence gates determine whether intervention-level actions are allowed. To test whether such a framework is actually useful, we introduce DEXPOSURE-BENCH; to show that it can be operated as an auditable system, we instantiate it as DEXPOSURE-CLAW.

Our contributions are:

- 1) **A forecast-grounded regulatory decision framework for DeFi exposure networks (§4).** We formulate DeFi risk supervision as a four-layer pipeline: FM prediction, deterministic monitoring and scenario analysis, decision

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policy, and data-health/confidence gating. This separates the problem of forecasting exposure dynamics from the problem of deciding when a supervisory ticket is warranted. Five network risk metrics with rolling baselines, five standardised stress scenarios with CVaR aggregation, and a four-level playbook make the framework auditable rather than a black-box LLM decision loop. While prior agentic work in DeFi targets transaction verification [1] or intent mining [2], and graph-based foundation models for risk have been explored only in traditional finance [3], no existing framework unifies forecasting, stress evidence, decision recommendation, and fail-safe gating for DeFi exposure networks.

- 2) **DEXPOSURE-BENCH: a benchmark for evaluating forecast-grounded risk-decision pipelines** (§5). To test the framework, we release a standardised harness that scores a candidate pipeline on six axes: forecasting quality, streaming anomaly detection, predictive uncertainty, what-if scenario fidelity, supervisory decision quality, and data-quality robustness. The bench operates on the full DEXPOSURE corpus (4,300 protocols, 24,300 tokens, 43.7M exposure entries, 283 weekly snapshots from 2020 to 2025 [4]), uses a regulator-aligned ground truth defined by *absolute* weight loss rather than fractional change, and ships eight reference method implementations. Generic LLM-agent benchmarks (SWE-BENCH, AGENTBENCH, HELM) do not evaluate domain-grounded supervisory decisions, and temporal-graph benchmarks (TGB, OGB) stop at structural prediction; DEXPOSURE-BENCH fills this benchmark gap for agentic risk pipelines.
- 3) **DEXPOSURE-CLAW: a deployable agentic instantiation and empirical characterisation** (§6). We package the framework as an auditable agentic system that emits ticket-level evidence bundles and can be run under a weekly supervisory cadence. On DEXPOSURE-BENCH, the full FM+LLM+gate stack raises ticket F1 from 0.0076 (persistence+rules) to 0.0233 (+207%). A tier-above Claude Opus 4.8 judge with a cross-family panel (Gemini 2.5 Pro, GPT-5.5) preserves the ordering  $m_7 \geq m_6 > m_2$ ; under the Opus 4.7 decision LLM the false-intervention rate rises to  $\approx 0.44$ . Replacing the decision LLM with a tier-below Sonnet 4.6 *strictly dominates* Opus 4.7 on both  $m_6$  and  $m_7$ : F1 rises further to 0.0288 on  $m_7$ , FIR collapses to 0.000, and judge quality reaches 2.65/5, at  $\sim 5\times$  lower inference cost—the deployable headline operating point.

We will release the implementation and benchmark artifacts with the project repository.

## 2 RELATED WORKS

This section positions DEXPOSURE-CLAW against five neighbouring threads: financial-network contagion, temporal graph learning, foundation models for time series and graphs, LLM agents in finance and DeFi, and benchmark design.

### 2.1 Financial Network Risk and DeFi Contagion

Systemic-risk analysis in traditional finance has long treated the financial system as a network of balance-sheet obli-

gations, funding links, and correlated asset exposures. Clearing-network models formalise how defaults propagate through interbank liabilities [5], while subsequent work studies when dense connections stabilise the system and when they amplify shocks [6], [7]. DebtRank-style measures further emphasise that a node can be “too central to fail” even when it has not yet defaulted [8]. These models motivate our focus on network position, concentration, and stress-test loss rather than protocol-level indicators alone.

DeFi changes the object being measured. Smart contracts replace part of the legal and operational layer, but composability, shared collateral, stablecoins, bridges, and liquidations create new contagion channels [9], [10], [11]. Recent work measures DeFi risk from aggregate deposits and borrowings [12], maps microscopic and systemic mechanisms across TradFi and DeFi [13], and studies fragility in TVL dynamics [14]. In contrast, our setting is an inter-protocol credit-exposure network: the units of analysis are weighted links between protocols, and the downstream task is not only to detect stress but also to produce traceable supervisory tickets.

### 2.2 Temporal Graph Learning for Dynamic Risk

Graph neural networks have become a natural tool for financial networks because they use both node attributes and relational structure. Prior work shows that GNNs can improve systemic-importance classification and systemic risk computation in simulated or interbank-style financial networks [15], [16], building on general graph convolution and attention architectures [17], [18]. These studies support the premise that risk is partly encoded in graph topology, not just in independent node features.

Our problem is additionally temporal: exposure weights, active protocols, and contagion paths evolve week by week. Dynamic-graph models such as DySAT, TGAT, EvolveGCN, and Temporal Graph Networks model changing neighbourhoods, temporal attention, recurrent graph filters, or event streams [19], [20], [21], [22]. Temporal graph benchmarks standardise link-prediction and node-classification tasks on dynamic graphs [23]. However, these settings typically optimise task-specific structural prediction. DEXPOSURE-CLAW instead uses temporal graph forecasts as inputs to monitoring, scenario stress tests, and LLM-based decisions; the forecasting layer is therefore evaluated by its effect on calibrated risk signals and recommendation tickets, not only by edge-level accuracy.

### 2.3 Foundation Models for Time Series and Graphs

Foundation models aim to replace narrowly trained task models with reusable representations learned from broad pretraining data [24]. Earlier neural forecasting architectures such as Temporal Fusion Transformers, Informer, and Autoformer established multi-horizon attention-based forecasting for long and heterogeneous time series [25], [26], [27]. More recently, Chronos, Lag-Llama, and TimesFM show that large pretrained sequence models can transfer across forecasting tasks and provide probabilistic forecasts [28], [29], [30]. In finance and economics, LLMs have also been used to extract signals from news and policy text for nowcasting pipelines [31], [32], [33].

Graph foundation modelling is less settled. Graphormer demonstrates that transformers can be made graph-aware through structural encodings [34]; self-supervised graph representation learning methods such as Deep Graph Info-max, GraphCL, and GraphMAE pretrain encoders through mutual information, contrastive augmentations, or masked attribute reconstruction [35], [36], [37]; and GraphPFN and TabPFN-style work explores how pretrained tabular predictors can transfer to graph-structured tasks [38], [39]. Heterogeneous graph pretraining has also been applied to default-risk propagation among bond issuers [3]. These systems motivate a foundation-model backbone for risk forecasting, but they do not directly address weighted temporal DeFi exposure graphs or the downstream requirements of uncertainty-gated supervisory decision making.

## 2.4 LLM Agents in Finance and DeFi

LLM agents are increasingly used for financial analysis, trading, and blockchain security. General agent patterns such as ReAct combine reasoning traces with tool-using actions [40], while financial LLM work such as FinGPT adapts language models to financial data and tasks [41]. In crypto portfolio management, multi-agent LLM systems coordinate market analysis and allocation decisions [42], and BlindTrade combines graph encoders with LLM agents for anonymised stock portfolio optimisation [43]. These systems are closest to ours in combining graph representations with language-model reasoning, but their objective is portfolio construction rather than systemic-risk supervision.

Within DeFi, agentic systems have focused mainly on security auditing, transaction understanding, and manipulation detection. Arbiter uses Graph-of-Thoughts reasoning for intent-transaction alignment [1]; Know Your Intent mines transaction intent with a multi-perspective LLM agent [2]; Heimdallr and Knowdit target smart-contract vulnerability detection [44], [45]; and PMDetector applies LLM reasoning to price manipulation detection [46]. Watson et al. combine graph anomaly detection with LLM-generated explanations for Bitcoin fraud [47]. These works show that LLMs can help explain or audit crypto systems, but they do not couple the agent with a dedicated graph time-series forecaster for network-level exposure risk.

## 2.5 Benchmarks for Graph, Forecasting, and Agentic Systems

The benchmark literature provides useful pieces but not the complete evaluation target required here. OGB and TGB standardise graph and temporal graph evaluation [23], [48]; time-series forecasting work emphasises temporally valid splits and robust model selection [49]; and general LLM-agent benchmarks such as HELM, SWE-bench, and AgentBench evaluate language capabilities, software repair, or generic agent behaviour [50], [51], [52].

DEXPOSURE-BENCH occupies a different point in this space. It evaluates forecasting, early warning, calibration, scenario stress fidelity, decision quality, and data-quality robustness in one harness, with ground truth tied to absolute exposure loss. This lets us ask whether a foundation-model forecast actually improves downstream risk monitoring and ticket generation, and also whether safety gates reduce

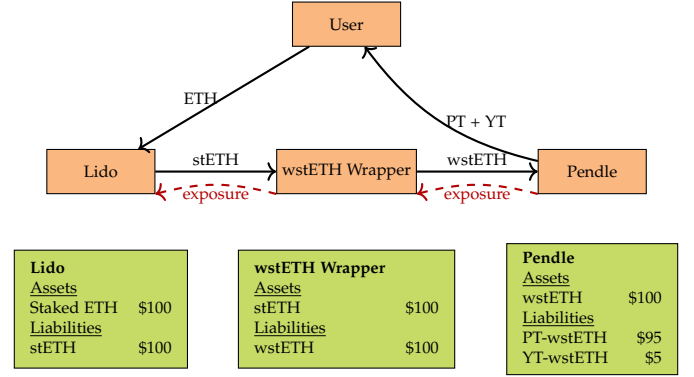


Fig. 1. Multi-layer credit exposure in DeFi. A user stakes ETH with Lido, receiving stETH, which is wrapped into wstETH and deposited into Pendle. Each protocol holds the liability of the preceding one, creating a chain of credit exposures from Pendle to Lido.

harmful recommendations under missingness or weak evidence.

## 3 NOTATION AND PRELIMINARIES

This section introduces the key terminology and notation used throughout the paper.

### 3.1 Credit Exposure in DeFi Networks

Let  $X_G(q)$  be the set of tokens issued (or generated) by protocol  $q$ , and let  $X_p(t)$  be the set of tokens held (or locked) by protocol  $p$  at time  $t$ . We say that protocol  $p$  has *credit exposure* to protocol  $q$  at time  $t$  if  $p$  holds at least one token issued by  $q$ :

$$X_p(t) \cap X_G(q) \neq \emptyset. \quad (1)$$

Equivalently,  $p$  is exposed to liabilities of  $q$ : if tokens issued by  $q$  devalue or become illiquid,  $p$  incurs losses. This token-mediated dependence generalises counterparty credit risk from traditional finance to DeFi, where exposures emerge through tokenised claims rather than explicit bilateral contracts. In practice, these dynamics can be tracked by reconstructing protocol-level TVL time series and monitoring how token composition changes over time [4].

Figure 1 shows how a single workflow can generate layered exposures. A user deposits ETH into Lido and receives stETH; stETH is wrapped into wstETH; wstETH is then deposited into Pendle, which splits it into PT-wstETH and YT-wstETH. On balance sheets, each step turns the upstream token into the downstream protocol's asset, so exposures stack: Pendle is exposed to the wstETH wrapper, which is exposed to Lido. If Lido suffers slashing or stETH de-pegs from ETH, losses can propagate through wstETH into Pendle and its PT/YT holders. This illustrates why DeFi exposures are naturally multi-layered and why DeXposure aims to measure them.

### 3.2 Mapping Tokens to Issuing Protocols

To implement Eq. (1), we must identify the issuing (or managing) protocol for each token. Let  $\mathcal{P}$  be the set of protocols and  $\mathcal{C}$  the set of blockchains. For any token  $\sigma$  appearing in protocol  $p$ , a *mapping function*  $M(\sigma)$  returns

the protocol  $q$  that issues or governs  $\sigma$ . We construct  $M$  via the following fallback pipeline:

- 1) *Metadata lookup*: match tokens to issuing protocols using DefiLlama metadata when available [53].
- 2) *Manual mapping*: for high-TVL tokens without metadata links, assign issuers using documentation and expert judgment.
- 3) *Text-vector similarity*: otherwise, vectorise token and protocol descriptions using TF-IDF [54] and assign  $M(\sigma)$  as the most similar protocol, provided the cosine similarity exceeds a threshold  $\theta$ .
- 4) *Primary market tokens*: if no match is obtained, treat the token as its own protocol (e.g., WETH).

This procedure attributes token movements to the appropriate issuer even when tokens are bridged or wrapped, and is used in curating DeXposure.

### 3.3 Network Representation and Value Flows

Over a discrete interval  $\tau = t_2 - t_1$ , we represent exposures by a weighted directed graph  $G_\tau = (P_\tau, E_\tau)$ , where  $P_\tau$  is the set of active protocols and  $E_\tau$  the directed edge set. Each node  $p \in P_\tau$  receives a *node weight* equal to the value of its held assets at  $t_2$ :

$$w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2, \sigma}, \quad (2)$$

where  $v_{t_2, \sigma}$  is the USD value of token  $\sigma$  at time  $t_2$ . Nodes below a threshold  $\theta$  may be pruned.

To define the *edge weight* from  $p$  to  $q$ , we first compute token-level value flows:

$$F_{pq, \tau}^\sigma = \begin{cases} \max(0, -\Delta S_{p, \tau}^\sigma), & \text{if } \Delta S_{p, \tau}^\sigma < 0, \\ \max(0, \Delta S_{q, \tau}^\sigma), & \text{if } \Delta S_{q, \tau}^\sigma \geq 0, \end{cases} \quad (3)$$

where  $\Delta S_{p, \tau}^\sigma = v_{p, t_2}^\sigma - v_{p, t_1}^\sigma$  is the change in the USD value of token  $\sigma$  held by  $p$  over  $\tau$ . Intuitively, if  $p$  reduces holdings while  $q$  increases holdings,  $\sigma$  is interpreted as flowing from  $p$  to  $q$ ; the construction enforces non-negativity.

The *edge weight*  $w_\tau(e_{pq})$  aggregates flows over tokens mapped to issuer  $q$ :

$$w_\tau(e_{pq}) = \sum_{\sigma \in X_{pq, \tau}} F_{pq, \tau}^\sigma, \quad (4)$$

where  $X_{pq, \tau}$  is the set of tokens issued by  $q$  that move from  $p$  to  $q$  over  $\tau$ . Positive  $w_\tau(e_{pq})$  corresponds to increasing exposure from  $p$  to  $q$ ; for single-token protocols, that token's flow equals the edge weight. In summary, we compute  $G_\tau$  by aggregating holdings at  $t_1$  and  $t_2$ , assigning node weights, pruning small nodes, and summing positive token flows into issuer-specific edge weights. The resulting graph sequence captures the evolving credit-dependency structure curated in DeXposure.

## 4 ALGORITHM

This section specifies the agentic system built on DEXPOSURE-FM for two tasks: network risk monitoring (Section 4.3–4.4) and decision recommendation (Section 4.5).

Figure 2 provides an architectural overview of the full system.

### 4.1 System Interface

At each decision epoch  $t$ , the agent observes a credit-exposure snapshot  $G_t = (V, E_t, W_t)$  and node covariates  $X_t$ , and interacts with DEXPOSURE-FM through a single forecast primitive:

$$\mathcal{P}_{t, h} \leftarrow \text{DEXPOSURE-FM.Forecast}(G_t, X_t, h), \quad \forall h \in \mathcal{H}, \quad (5)$$

where  $\mathcal{P}_{t, h}$  denotes the model's predictive distribution over the future network  $G_{t+h}$ , and  $\mathcal{H} = \{1, 4, 8, 12\}$  (weeks). From  $\mathcal{P}_{t, h}$  we construct a representative predicted graph  $\hat{G}_{t+h}$  using a thresholded expected-weight operator:

$$\hat{w}_{pq, t+h} := \Pr(e_{pq, t+h} = 1) \cdot \mathbb{E}[w_{pq, t+h} \mid e_{pq, t+h} = 1], \quad (6)$$

$$\hat{E}_{t+h} := \{(p, q) : \Pr(e_{pq, t+h} = 1) > \pi_{\min}\}. \quad (7)$$

where  $\pi_{\min}$  (default 0.2, tuned on validation) balances edge coverage and sparsity. For uncertainty quantification, we draw  $M$  Monte Carlo samples  $\{\tilde{G}_{t+h}^{(m)}\}_{m=1}^M$  by applying Bernoulli edge sampling and Gaussian weight perturbation to  $\mathcal{P}_{t, h}$  (default  $M = 50$ ).

The agent returns three outputs: (a) a set of monitoring alerts with evidence and confidence, (b) a scenario stress summary, and (c) a ranked list of recommendation tickets.

### 4.2 Data-Health Gating

Real-world DeFi graphs can be sparse, stale, or corrupted. Before producing actionable outputs, the agent computes a scalar data-health score from four deterministic checks—freshness, feature missingness, topology sanity (edge-to-node ratio), and discontinuity detection:

$$\begin{aligned} DH_t &:= \text{mean}(\text{fresh}_t, \text{miss}_t, \text{topo}_t, \text{disc}_t) \in [0, 1], \\ \text{SAFE\_MODE}_t &:= \mathbb{I}\{DH_t < \tau_{\text{data}}\}. \end{aligned} \quad (8)$$

When  $DH_t$  falls below  $\tau_{\text{data}}$  (default 0.7), the system enters safe mode: monitoring alerts are still emitted, but intervention-type recommendations are suppressed (Section 4.5).

### 4.3 Task I: Network Risk Monitoring

Monitoring metrics ( $\Phi$ ): Risk in exposure networks manifests as concentration (a few protocols dominate), fragility (shocks propagate widely), and instability (structure shifts rapidly). To capture these dimensions, the monitoring agent evaluates five functionals on  $\hat{G}_{t+h}$ :

Formally, at each horizon  $h$  we compute:

$$\mathbf{N}_{t, h} := \Phi(\hat{G}_{t+h}, X_t) \in \mathbb{R}^5, \quad \tilde{\mathbf{N}}_{t, h}^{(m)} := \Phi(\tilde{G}_{t+h}^{(m)}, X_t). \quad (9)$$

Alert generation: An alert fires when a predicted metric departs from its rolling historical baseline by more than  $z$  standard deviations (default  $z = 2$ , baseline window  $L = 26$  weeks):

$$\mathcal{A}_t := \left\{ a = (h, j) : \frac{|N_{t, h}^{(j)} - \mu_t^{(j)}|}{\max(\sigma_t^{(j)}, \epsilon)} > z \right\}, \quad (10)$$

where  $(\mu_t^{(j)}, \sigma_t^{(j)})$  are the rolling mean and standard deviation of metric  $j$  over the last  $L$  observed snapshots, and  $\epsilon = 10^{-8}$  prevents division by zero.

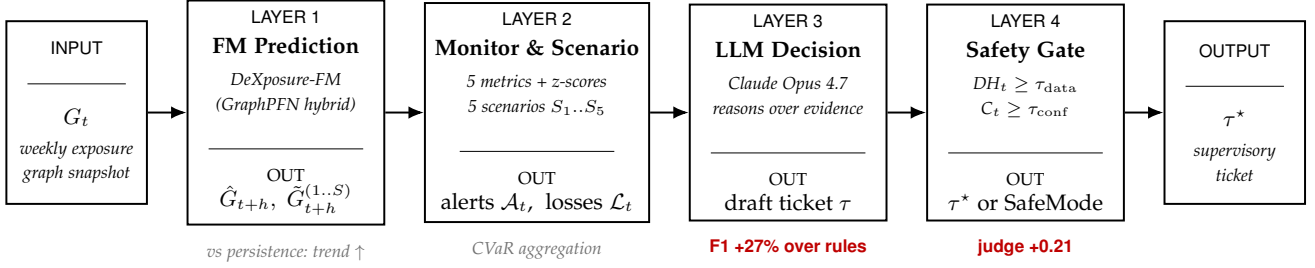


Fig. 2. DEXPOSURE-CLAW: a four-layer query architecture. The system ingests the weekly exposure graph  $G_t$ , runs FM prediction (Layer 1), monitoring and scenario engines (Layer 2), an LLM decision layer (Layer 3), and a safety gate (Layer 4) that emits either the supervisory ticket  $\tau^*$  or a safe-mode notice.

| ID | Metric              | Definition                                                  |
|----|---------------------|-------------------------------------------------------------|
| N1 | Systemic Importance | $\max_{p \in V} \text{PageRank}(\hat{G}_{t+h})_p$           |
| N2 | HHI Concentration   | $\sum_p (d_p / \sum_q d_q)^2$ , $d_p$ = weighted out-degree |
| N3 | Network Density     | $ E_{t+h}  / ( V ( V  - 1))$                                |
| N4 | PageRank Gini       | $\text{Gini}(\{\text{PR}_p\}_{p \in V})$                    |
| N5 | Degree Gini         | $\text{Gini}(\{d_p\}_{p \in V})$                            |

TABLE 1

Monitoring functionals  $\Phi$ . PageRank uses power iteration with damping 0.85 and 10 iterations. Gini coefficient:

$$1 - 2 \sum_{i=1}^n \sum_{k=1}^i x_{(k)} / (n \cdot \sum_k x_{(k)}), \text{ where } x_{(k)} \text{ are sorted values.}$$

| ID | Scenario                | Target                    | Shock |
|----|-------------------------|---------------------------|-------|
| S1 | Single protocol failure | Highest-weight node       | 100%  |
| S2 | Bridge cluster failure  | All bridge-category nodes | 100%  |
| S3 | Stablecoin de-peg       | All stablecoin nodes      | 50%   |
| S4 | Sector-wide shock       | All lending nodes         | 30%   |
| S5 | Correlated stress       | Top-10 nodes by weight    | 20%   |

TABLE 2

Standardised stress-scenario library  $\mathcal{S}_0$ .

**Evidence and attribution:** Each alert  $a \in \mathcal{A}_t$  carries an evidence bundle  $\text{Ev}_t(a) = (N_{t,h}^{(j)}, \mu_t^{(j)}, \sigma_t^{(j)}, \text{Attr}_t(a), h)$ , where  $\text{Attr}_t(a)$  is a ranked list of top- $K$  contributing nodes/edges obtained via marginal contribution approximation:  $\text{Contrib}(e) := \Phi^{(j)}(\hat{G}_{t+h}) - \Phi^{(j)}(\hat{G}_{t+h} \setminus e)$  (default  $K = 10$ ).

**Uncertainty-aware confidence:** Alert reliability depends on three factors: data quality, forecast dispersion, and horizon length. Using Monte Carlo samples, we compute the coefficient of variation  $\text{CV}_{t,h}^{(j)} := \sigma(\{\tilde{N}_{t,h}^{(m,j)}\}_m) / \bar{N}_{t,h}^{(j)}$  (clamped to  $[0, 1]$ ) and define:

$$C_t(a) := DH_t \cdot \frac{1}{1 + \text{CV}_{t,h}^{(j)}} \cdot \frac{1}{1 + \ln(1 + h)} \in [0, 1]. \quad (11)$$

This score monotonically decreases with lower data health, wider predictive spread, and longer horizons, ensuring that unreliable alerts are automatically downweighted.

#### 4.4 Scenario Engine

To quantify tail risk beyond metric-level alerts, the agent evaluates a standardised stress-scenario library  $\mathcal{S}_0$ . Each scenario applies a deterministic shock—removing or reducing edge weights of targeted nodes—and propagates losses through the network:

For each scenario  $s$  and horizon  $h$ , we compute the systemic loss on both the predicted graph and each MC sample:

$$L_{t,h}(s) := \frac{\sum_e w_e(\hat{G}_{t+h}) - \sum_e w_e(\hat{G}_{t+h}^s)}{\sum_e w_e(\hat{G}_{t+h})}, \quad (12)$$

$$\tilde{L}_{t,h}^{(m)}(s) := \ell(\tilde{G}_{t+h}^{(m)}; s). \quad (13)$$

where  $\hat{G}_{t+h}^s$  denotes the network after applying shock  $s$ . The scenario engine also tracks the number of *distressed* nodes (those losing  $> 50\%$  of total edge weight) and a propagation depth estimate  $\lceil \log_2(\text{affected nodes} + 1) \rceil + 1$ .

**CVaR aggregation:** To capture tail risk, we compute an empirical CVaR from the MC sample losses. Let  $\lambda \in [0, 1]$  (default 0.5) control the tail fraction. We sort the  $M$  sample losses in descending order and average the worst  $\lceil \lambda M \rceil$ :

$$\text{CVaR}_\lambda(s, h) := \frac{1}{\lceil \lambda M \rceil} \sum_{i=1}^{\lceil \lambda M \rceil} \tilde{L}_{t,h}^{(i)}(s), \quad (14)$$

where  $\tilde{L}^{(1)} \geq \tilde{L}^{(2)} \geq \dots$  are order statistics. The scenario summary  $\mathcal{R}_t$  ranks scenarios by  $L_{t,h}(s)$  and retains the worst-case scenario across all horizons.

#### 4.5 Task II: Decision Agent

**Action space:** The decision module maps alerts and scenario results to recommendation tickets using a conservative playbook with four action types, ordered by escalation severity:

**Feasibility constraints:** An intervention-type action  $u$  (those requiring safe-mode off) is feasible only when both the data-health gate and the confidence gate are satisfied:

$$u \in \mathcal{C}_t \iff (\text{SAFE\_MODE}_t = 0) \wedge (\bar{C}_t \geq \tau_{\text{conf}}), \quad (15)$$

| Action               | Severity | Weight $w_u$ | Safe-mode OK? | Min. alerts |
|----------------------|----------|--------------|---------------|-------------|
| Monitor              | Low      | 0.25         | Yes           | 1           |
| Investigate          | Medium   | 0.50         | Yes           | 1           |
| Recommend<br>-Reduce | High     | 0.75         | No            | 2           |
| Contingency          | Critical | 1.00         | No            | 3           |

TABLE 3  
Playbook action space  $\mathcal{U}$  with severity weights and feasibility constraints.

where  $\bar{C}_t = \frac{1}{|\mathcal{A}_t|} \sum_{a \in \mathcal{A}_t} C_t(a)$  is the mean alert confidence and  $\tau_{\text{conf}}$  (default 0.6) gates interventions. Non-interventional actions (Monitor, Investigate) are always feasible when at least one alert exists.

**Ticket scoring:** Each feasible recommendation is scored by a multiplicative heuristic combining playbook severity, alert confidence, and scenario impact:

$$\text{Score}_t(u) := w_u \cdot \bar{C}_t \cdot \text{Imp}_t, \quad (16)$$

where  $w_u$  is the severity weight (Table 3),  $\bar{C}_t$  is the mean alert confidence, and  $\text{Imp}_t := \frac{1}{|\mathcal{S}_0|} \sum_{s \in \mathcal{S}_0} \text{CVaR}_\lambda(s)$  aggregates scenario impact as the mean CVaR across all scenarios. When no scenarios are available,  $\text{Imp}_t$  defaults to 1.0 (neutral multiplier), ensuring alert-only tickets receive non-zero scores. The agent returns the top- $N$  tickets (default  $N = 10$ ) ranked by  $\text{Score}_t(u)$ .

**Target selection:** For intervention tickets, targets are selected by collecting the top- $K$  attributed nodes from all triggering alerts and worst-case scenarios, ranked by frequency-weighted vote.

## 4.6 End-to-End Procedure

Algorithm 1 summarises the full agent loop.

## 5 DEXPOSURE-BENCH: AN EVALUATION SUITE FOR FM+LLM DECISION PIPELINES

### 5.1 Motivation: an evaluation gap

Foundation-model pipelines that pair a domain forecaster with a large language model decision layer are proliferating across finance, supply-chain risk, and infrastructure monitoring. Despite this, the community lacks a standardised suite for evaluating the full *pipeline*—most existing benchmarks evaluate a single stage in isolation. Generic LLM-agent benchmarks (SWE-BENCH [51], AGENTBENCH [52], HELM [50]) cover open-ended coding or tool use but not domain-grounded supervisory decisions. Temporal-graph benchmarks (TGB [23], OGB [48]) measure structural prediction quality but stop short of any decision metric. Domain-specific evaluations of DeFi risk forecasting exist as one-off experiments accompanying single papers, none of them packaged as a reusable harness.

DEXPOSURE-BENCH fills this gap with a single, reproducible suite that scores a candidate pipeline on six axes spanning forecast quality, anomaly detection, predictive uncertainty, what-if query fidelity, downstream decision quality, and data-quality robustness, on a shared 16,610-protocol DeFi exposure dataset with a *regulator-aligned* ground truth.

### Algorithm 1 DeXposure-Claw: one epoch

**Require:** State  $(G_t, X_t)$ ; horizons  $\mathcal{H}$ ; scenarios  $\mathcal{S}_0$ ; thresholds  $(\tau_{\text{data}}, \tau_{\text{conf}}, z, \pi_{\text{min}})$ ; MC count  $M$   
**Ensure:** Alerts  $\mathcal{A}_t$ ; scenario summary  $\mathcal{R}_t$ ; tickets  $\mathcal{D}_t$

- 1:  $DH_t \leftarrow \text{DataHealth}(G_t, X_t)$ ;  $\text{SAFE\_MODE}_t \leftarrow \mathbb{I}\{DH_t < \tau_{\text{data}}\}$
- 2: **for all**  $h \in \mathcal{H}$  **do**
- 3:  $\mathcal{P}_{t,h} \leftarrow \text{DEXPOSURE-FM.Forecast}(G_t, X_t, h)$
- 4:  $\hat{G}_{t+h} \leftarrow \text{BuildPredGraph}(\mathcal{P}_{t,h}; \pi_{\text{min}})$   $\triangleright$  Eq. (6)
- 5:  $\{\tilde{G}_{t+h}^{(m)}\}_{m=1}^M \leftarrow \text{MCSample}(\mathcal{P}_{t,h})$   $\triangleright$  Bernoulli edges + Gaussian weights
- 6:  $\mathbf{N}_{t,h} \leftarrow \Phi(\hat{G}_{t+h}, X_t)$ ;  $\tilde{\mathbf{N}}_{t,h}^{(m)} \leftarrow \Phi(\tilde{G}_{t+h}^{(m)}, X_t)$   $\triangleright$  Table 1;  $\mathbf{N}_1, \dots, \mathbf{N}_5$
- 7:  $(\mu_t^{(j)}, \sigma_t^{(j)}) \leftarrow \text{RollingStats}(\{\mathbf{N}_{t-l,h}\}_{l=1}^L)$   $\triangleright L = 26$  weeks
- 8:  $\mathcal{A}_{t,h} \leftarrow \{(h, j) : |\mathbf{N}_{t,h}^{(j)} - \mu_t^{(j)}| / \max(\sigma_t^{(j)}, \epsilon) > z\}$   $\triangleright$  Eq. (10)
- 9: Attach  $\text{Evt}_t(a)$  and  $C_t(a)$  for all  $a \in \mathcal{A}_{t,h}$   $\triangleright$  Eq. (11), uses  $\tilde{\mathbf{N}}_{t,h}^{(m)}$  for CV
- 10: **for all**  $s \in \mathcal{S}_0$  **do**
- 11:  $L_{t,h}(s) \leftarrow \text{StressLoss}(\hat{G}_{t+h}, s)$ ;  $\tilde{L}_{t,h}^{(m)}(s) \leftarrow \text{StressLoss}(\tilde{G}_{t+h}^{(m)}, s)$
- 12:  $\text{CVaR}_\lambda(s, h) \leftarrow \text{TailMean}(\{\tilde{L}_{t,h}^{(m)}(s)\}_{m=1}^M, \lambda)$   $\triangleright$  Eq. (14)
- 13: **end for**
- 14: **end for**
- 15:  $\mathcal{A}_t \leftarrow \bigcup_{h \in \mathcal{H}} \mathcal{A}_{t,h}$
- 16:  $\mathcal{R}_t \leftarrow \text{RankScenarios}(\{L_{t,h}(s), \text{CVaR}_\lambda(s, h)\}_{h,s})$
- 17:  $\mathcal{D}_t \leftarrow \text{PlaybookDecision}(\mathcal{A}_t, \mathcal{R}_t, DH_t, \text{SAFE\_MODE}_t, \tau_{\text{conf}})$   $\triangleright$  Table 3
- 18: **return**  $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$

### 5.2 Suite schema

The suite is organised as six benchmarks, each producing an independently reportable scalar (or vector of scalars). Pipelines may report against any subset; the full leaderboard requires all six. Table 4 summarises each axis.

The six axes were chosen so that they are: (i) *non-overlapping* in what they measure—a system can excel at b1 while failing b5; (ii) *decomposable along the four-layer framework* (§4), so that ablations on the predictor layer (b1, b3, b6), monitor layer (b2), scenario layer (b4), and decision layer (b5) can be reported independently; and (iii) *individually publishable*, so that downstream work may extend a single axis without re-running the whole suite.

### 5.3 Dataset and split

The benchmark draws on the DEXPOSURE weekly exposure-graph dataset [4], a directed weighted graph  $G_t = (V_t, E_t, w_t)$  at each weekly timestep  $t$ , with  $V_t$  a subset of the 16,610-protocol universe observed over 2020-03–2025-08 (283 weeks). Node features carry log-TVL, number of token types, maximum token share, Shannon entropy of the token distribution, and a 15-class category one-hot. Edge weights are log-scaled USD inter-protocol exposures. We adopt the strict no-leakage temporal split of Table 5.

TABLE 4  
DEXPOSURE-BENCH schema. Each row is an independently reportable benchmark; the full leaderboard requires all six.

| ID             | Capability tested                 | Primary metrics                                                                                           | Applicable methods          |
|----------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------|
| b1_forecast    | Temporal graph prediction quality | PageRank MAE, HHI MAE, Gini MAE, trend consistency, rank correlation; per horizon $h \in \{1, 4, 8, 12\}$ | Any predictor               |
| b2_warning     | Streaming anomaly detection       | Precision, recall, F1, lead time, stability; per event $\times$ alert budget                              | Any monitor + predictor     |
| b3_calibration | Predictive uncertainty quality    | PI coverage @ 0.90 target, PI width, ECE, CRPS                                                            | Predictors with intervals   |
| b4_stress      | What-if scenario fidelity         | Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@ $k$ ; per scenario $S_1 \dots S_5$ | Predictor + scenario engine |
| b5_decision    | Supervisory ticket quality        | Precision, recall@ $k$ , F1, target stability, judge score, FIR                                           | Predictor + decision policy |
| b6_robustness  | Data quality sensitivity          | Per-benchmark metrics under 5 degradation regimes; relative degradation                                   | Any predictor               |

TABLE 5  
DEXPOSURE-BENCH canonical split. All reported numbers in this paper come from the test split.

| Split      | Period          | Weeks | Usage                                   |
|------------|-----------------|-------|-----------------------------------------|
| Train      | 2020-03–2024-06 | ~220  | Predictor training                      |
| Validation | 2024-07–2024-12 | ~26   | Conformal calibration, threshold tuning |
| Test       | 2025-01–2025-08 | ~33   | Frozen leaderboard                      |

#### 5.4 Ground truth: regulator-aligned absolute loss

Prior systemic-risk evaluations rank protocols by *fractional* weight change, which produces a ground-truth pool dominated by small protocols whose deterioration carries low systemic significance. This choice silently biases every downstream metric and is not what supervisory authorities monitor in practice.

DEXPOSURE-BENCH therefore defines its ground truth in terms of *absolute* exposure loss. Let  $w_t(v) = \sum_{e \ni v} w_t(e)$  be the total weight incident on protocol  $v$  at time  $t$ . For horizon  $h$  and lookahead window  $W$  we define the *stressed set*  $S_t^h$  as the top  $\pi$  fraction of active protocols ranked by absolute loss:

$$\begin{aligned} \Delta_t^h(v) &= w_t(v) - w_{t+h}(v), \\ S_t^h &= \text{top}_\pi \{v : w_t(v) > 0, \Delta_t^h(v) > 0\}. \end{aligned} \quad (17)$$

We fix  $\pi = 0.05$  throughout (yielding an average pool of  $|S_t^h| \approx 283$  protocols per week, stable across the test split) and the lookahead window  $W = 4$  weeks.

Equation 17 is the single ground-truth definition used by b2, b5, and the LLM-decision evaluation pipeline. Aligning the metric with absolute economic magnitude—rather than fractional change—restores discrimination between methods that target central versus peripheral nodes and matches how a supervisor would prioritise.

#### 5.5 Methods supported out of the box

Eight reference methods are bundled with the harness, sufficient to populate a complete leaderboard from a fresh checkout. Each method exposes the same `predict_graph` or `decide_actions` interface, so adding a ninth method is a single-file change.

The eight listed in Table 6 together expose all four cuts of the framework ablation: (i) adding the predictor layer (m1→m5), (ii) swapping the decision layer (m5→m6), (iii) adding

the safety gate (m6→m7), and (iv) removing the predictor from the LLM stack (m2←m6).

#### 5.6 Reporting protocol

To enter the leaderboard, a submission produces one JSON file per {benchmark, method} pair following the harness’s published schema, and a single `results.json` aggregating the suite. Required reports for the headline table are b1, b3, b5, and b6; b2 and b4 are required only for systems that implement a monitor or scenario layer respectively. All metrics are computed on the frozen test split of Table 5; training and hyperparameter tuning may use train + validation freely. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files so that any reported number can be reproduced bit-exactly.

#### 5.7 Positioning in the benchmark landscape

Table 7 compares DEXPOSURE-BENCH against the closest existing evaluation suites. No prior suite covers a domain-grounded decision metric on top of a forecasting backbone, which is the niche DEXPOSURE-BENCH addresses.

#### 5.8 Reproducibility and extensibility

The harness, dataset, and reference implementations are released under an open licence at the project repository. A complete leaderboard run, including all eight reference methods and the LLM evaluation pipeline, completes in under four GPU-hours on a single RTX 4090 plus approximately ten USD of LLM API spend, making the benchmark accessible to academic groups without industrial compute. Extending the suite—adding a method, an additional ground-truth formulation, or a new benchmark axis—is supported by single-file plug-in points documented alongside the harness.

### 6 EXPERIMENTS

We use DEXPOSURE-BENCH (§5) to evaluate whether the proposed forecast-grounded decision framework works when instantiated as DEXPOSURE-CLAW. The experiments answer three questions. First, does the forecasting layer produce risk signals that are useful for supervision, even when persistence remains strong on static point metrics? Second, do FM and LLM decision layers improve supervisory tickets



TABLE 6  
Reference methods bundled with DEXPOSURE-BENCH. Heuristic identifiers use the  $h$  namespace; learned methods use  $m$ .

| ID                   | Predictor       | Policy            | Role                       |
|----------------------|-----------------|-------------------|----------------------------|
| h1_weighted_degree   | —               | heuristic alert   | Streaming anomaly baseline |
| m1_persistence_rules | $G_{t+h} = G_t$ | rule engine       | Decision baseline          |
| m2_snapshot_llm      | current snap    | LLM               | LLM-only ablation          |
| m3_evolvegcnn        | EvolveGCN       | —                 | Learned-GNN baseline       |
| m4_fm_only           | FM              | —                 | FM-only ablation           |
| m5_fm_rules          | FM              | rule engine       | FM + rules                 |
| m6_fm_llm            | FM              | LLM               | Full FM + LLM stack        |
| m7_fm_llm_gated      | FM              | LLM + safety gate | Safety-gated FM + LLM      |

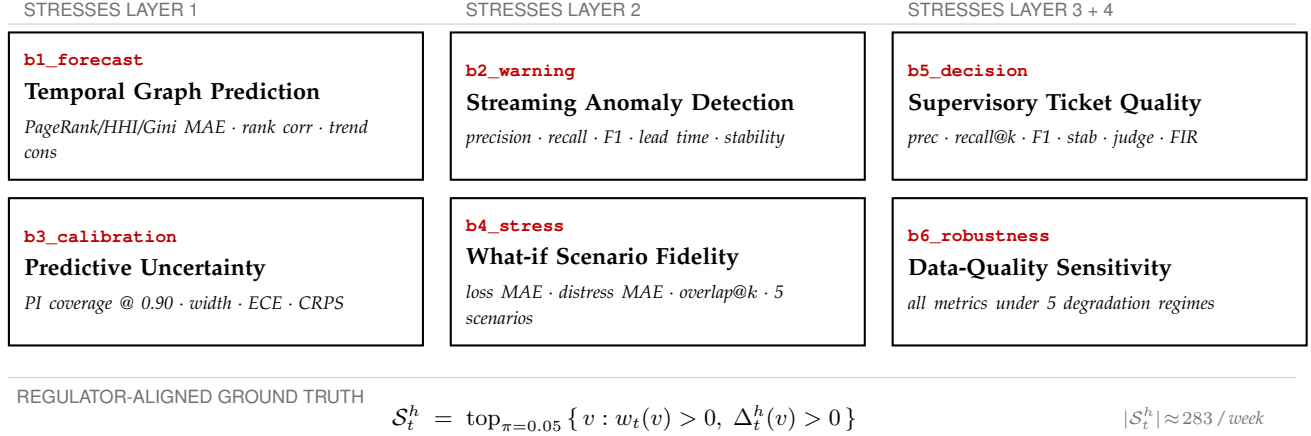


Fig. 3. DEXPOSURE-BENCH: a six-axis evaluation suite. The six benchmarks are grouped by which framework layer they stress; the bottom strip carries the regulator-aligned ground truth that anchors all decision-quality metrics.

TABLE 7  
DEXPOSURE-BENCH versus closest existing suites. A check mark indicates a covered axis; — indicates the axis is out of scope.

| Suite                       | Pred. | Anom. | Calib. | Scen. | Decision |
|-----------------------------|-------|-------|--------|-------|----------|
| OGB [48]                    | ✓     | —     | —      | —     | —        |
| TGB [23]                    | ✓     | —     | —      | —     | —        |
| HELM [50]                   | —     | —     | —      | —     | ✓        |
| SWE-BENCH [51]              | —     | —     | —      | —     | ✓        |
| AGENTBENCH [52]             | —     | —     | —      | —     | ✓        |
| DEXPOSURE-BENCH (this work) | ✓     | ✓     | ✓      | ✓     | ✓        |

once precision, recall, explanations, and false interventions are measured together? Third, do the gates, scenario engine, and multi-horizon design provide deployability evidence rather than ornamental system components? Task I (b1, b2, b3, b6) addresses the forecasting and monitoring question; Task II (b4, b5) addresses the decision-quality question; the ablations address deployability. Unless explicitly marked as a historical event study, results are reported on the frozen 2025 test split of Table 5; the ground truth, dataset split, and reference methods used throughout are defined in §5.

## 6.1 Task I: Network Risk Monitoring

### 6.1.1 Implementation Details

Data and temporal granularity: We use the DEXPOSURE dataset [4], comprising weekly directed weighted exposure-graph snapshots sourced from the DeFiLlama API. The dataset spans 2020-03 to 2025-08 (283 weeks), with

50–500 protocol nodes and 100–2,000 exposure edges per snapshot, totalling over 43.7M data entries. Node features include log-TVL, number of token types, maximum token share, Shannon entropy of token distribution, and protocol category (one-hot, 15 classes). Edge weights are log-scaled USD exposure values. We evaluate multi-horizon forecasting on  $\mathcal{H} = \{1, 4, 8, 12\}$  weeks.

Temporal split: We enforce a strict no-leakage temporal split:

| Split | Period            | Weeks | Usage                             |
|-------|-------------------|-------|-----------------------------------|
| Train | 2020-03 – 2024-06 | ~222  | FM backbone training              |
| Val   | 2024-07 – 2024-12 | ~26   | Hyperparameter + threshold tuning |
| Test  | 2025-01 – 2025-08 | ~33   | Final evaluation                  |

Stress event windows: For early-warning evaluation (b2\_warning), we additionally run a historical event study around three major DeFi crises that occur before the 2025 test split. This diagnostic evaluates whether the monitor turns before known shocks; it is reported separately from the predictor-specific 2025 leaderboard:

| Event      | Date       | Pre-shock (4w) | Event (3w)    |
|------------|------------|----------------|---------------|
| Terra/Luna | 2022-05-09 | 03-28 – 05-02  | 05-02 – 05-23 |
| FTX        | 2022-11-07 | 09-26 – 10-31  | 10-31 – 11-21 |
| SVB/USDC   | 2023-03-10 | 02-06 – 03-06  | 03-06 – 03-27 |

Forecast interface and predicted-graph construction: For each epoch  $t$  and horizon  $h \in \mathcal{H}$ , we query the backbone model to obtain a predictive distribution  $\mathcal{P}_{t,h}$ . We construct an expected-weight predicted graph  $\hat{G}_{t+h}$  by thresholding edge existence probabilities at  $\pi_{\min}$  and



setting  $\hat{w}_{pq} := \Pr(e_{pq} = 1) \cdot \mathbb{E}[w_{pq} \mid e_{pq} = 1]$  (Eq. (6)). We set  $\pi_{\min} = 0.2$  on the validation split. For uncertainty quantification, we draw  $M = 50$  Monte Carlo samples via Bernoulli edge sampling and Gaussian weight perturbation.

**Monitoring metrics and alert triggers:** At each horizon, we compute five monitoring functionals  $N_{t,h} = \Phi(\hat{G}_{t+h}, X_t)$  (Table 1): systemic importance (N1), HHI concentration (N2), network density (N3), PageRank Gini (N4), and degree Gini (N5). Baseline bands  $(\mu_t^{(j)}, \sigma_t^{(j)})$  are computed via a rolling window of  $L = 26$  weeks. An alert fires when a metric exceeds  $z = 2$  standard deviations from baseline (Eq. (10)). Alert confidence follows Eq. (11).

**Stress scenarios:** We evaluate five standardised scenarios  $S_0$  (Table 2): single protocol failure (S1, 100% shock on top node), bridge cluster failure (S2, 100% on bridges), stablecoin de-peg (S3, 50% on stablecoins), sector-wide lending shock (S4, 30%), and correlated stress (S5, 20% on top-10 nodes). CVaR is computed as the mean of the worst  $\lambda = 0.5$  fraction of MC sample losses (Eq. (14)).

**Attribution:** For each triggered alert, we rank contributing edges by marginal contribution:  $\text{Contrib}(e) := \Phi^{(j)}(\hat{G}_{t+h}) - \Phi^{(j)}(\hat{G}_{t+h} \setminus e)$ , retaining the top  $K = 10$ .

**Evaluation protocol:** Within the six-axis framework of Fig. 3, Task I evaluates the forecast-measure pipeline along three axes:

- 1) *Risk-metric forecasting (b1\_forecast)*: predicted vs. realised metrics using MAE and Spearman rank correlation, for each  $h \in \{1, 4, 8, 12\}$ .
- 2) *Early-warning quality (b2\_warning)*: lead time to stress events, precision/recall under alert budgets  $\{5, 10, 20\}$ , alert stability (flip rate), and F1-warning.
- 3) *Uncertainty calibration (b3\_calibration)*: expected calibration error (ECE), prediction interval coverage (target 90%), PI width, and CRPS. We apply split conformal prediction on the validation set to ensure asymptotically valid coverage.

**Reproducibility:** All hyperparameters ( $\pi_{\min}, L, z, K, M, \tau_{\text{data}}$ ) are selected on the validation split and fixed for test evaluation. Random seeds are fixed for MC sampling and LLM consistency checks. The pipeline is fail-closed on predictors: an unavailable FM or EvolveGCN checkpoint aborts the corresponding run rather than silently falling back to persistence. For *b2\_warning* we report the shared weighted-degree heuristic (*h1\_weighted\_degree*), since *b2*'s monitoring layer sits upstream of the predictor choice and is therefore not a method-specific comparison.

### 6.1.2 Empirical Results

Table 8 reports the Task I evidence ledger used in the main text; the full all-horizon *b1\_forecast* matrix is provided as Table 14 in Appendix A. The point of the table is not to claim that the FM wins every forecasting metric. Instead, it separates three types of evidence: 2025 predictor-specific evidence (*b1*, *b3*, *b6*), a historical monitor event study (*b2*), and the strong persistence baseline that any deployable monitor must acknowledge.

Persistence is a deceptively strong point-metric baseline: On PageRank MAE, Gini MAE, and rank correlation, persistence matches or marginally beats the FM at every horizon (rank correlation 0.570 vs 0.558 at  $h=4$ ; PR MAE

3.4 vs 4.5 in  $10^{-5}$  units). Weekly DeFi exposure structure is heavily auto-correlated, so any predictor that returns the most recent graph captures the dominant source of variance “for free”—the same phenomenon that makes seasonal-naïve a strong baseline in classical forecasting. A new model must clear this bar before its other properties matter.

FM gains are concentrated in trend and robustness, not in absolute error: The FM produces a non-trivial trend signal (trend consistency 0.525–0.632 across horizons) while persistence is structurally pinned at 0. Under data degradation (*b6\_robustness*, mean relative degradation 0.113 vs 0.148 for persistence), the FM drops 24% less on average. These results support a narrower claim than pointwise superiority: the FM contributes direction, uncertainty, and robustness features that persistence does not provide, while persistence remains the better static point-metric baseline.

Calibration is strong; learned GNN transfer remains difficult: The FM achieves PI coverage 0.913 against a 0.90 target with ECE 0.013 on *b3\_calibration*, both within the expected slack of split-conformal calibration. Persistence has no native uncertainty estimate and is omitted from this row. EvolveGCN is generally weaker than persistence and FM variants on the downstream metrics in the appendices, which we attribute to data efficiency: 222 weekly training snapshots fall below the regime where it learns a useful structural prior, while the FM benefits from large-scale pre-training transfer.

## 6.2 Task II: Decision-Making

### 6.2.1 Implementation Details

**Decision timeline and inputs:** Task II operates at the same decision epochs as Task I. At each epoch, the decision agent receives: (i) the current state  $(G_t, X_t)$ , (ii) multi-horizon forecasts  $\{\hat{G}_{t+h}\}_{h \in \mathcal{H}}$  and optional MC samples, (iii) monitoring alerts  $\mathcal{A}_t$  with evidence and confidence scores, and (iv) the scenario summary  $\mathcal{R}_t$ .

**Action space and playbook:** We use a four-level conservative playbook (Table 3): *Monitor* (severity weight 0.25, always feasible), *Investigate* (0.50, always feasible), *Recommend-Reduce* (0.75, requires  $\text{SAFE\_MODE} = 0$  and  $\geq 2$  alerts), and *Contingency* (1.00, requires  $\text{SAFE\_MODE} = 0$  and  $\geq 3$  alerts). Intervention-type actions additionally require mean alert confidence  $\bar{C}_t \geq \tau_{\text{conf}} = 0.6$ .

**Ticket scoring:** Tickets are ranked by  $\text{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \text{Imp}_t$  (Eq. (16)), where  $w_u$  is the playbook severity weight,  $\bar{C}_t$  is mean alert confidence, and  $\text{Imp}_t$  is the mean CVaR across all scenarios (defaulting to 1.0 when no scenarios are available). We output the top- $N = 10$  tickets.

**Target selection:** For intervention tickets, targets are selected from the union of top- $K$  attributed nodes across triggering alerts and worst-case scenarios, ranked by frequency-weighted vote.

**Evaluation protocol:** We evaluate Task II using both offline and scenario-based criteria:

- 1) *Offline (b5\_decision)*: ticket precision (fraction followed by real stress), target stability (Jaccard between adjacent epochs), audit completeness (evidence bundle coverage), suppression rate (safe-mode activation), and false intervention rate.

| Evidence item                         | Scope / method           | Value | Comparator    |
|---------------------------------------|--------------------------|-------|---------------|
| PR-MAE@h4 ( $10^{-5}$ , lower better) | 2025 b1, FM              | 4.5   | Persist 3.4   |
| RankCorr@h4                           | 2025 b1, FM              | 0.558 | Persist 0.570 |
| TrendCons@h4                          | 2025 b1, FM              | 0.628 | Persist 0.000 |
| HHI-MAE@h4, lower better              | 2025 b1, FM              | 0.032 | Persist 0.032 |
| Precision, mean                       | Historical b2, shared h1 | 0.756 | N/A           |
| F1-warning, mean                      | Historical b2, shared h1 | 0.007 | N/A           |
| Lead time, weeks                      | Historical b2, shared h1 | 4.667 | N/A           |
| ECE                                   | 2025 b3, FM intervals    | 0.013 | N/A           |
| PI coverage                           | 2025 b3, FM intervals    | 0.913 | target 0.900  |
| Relative degradation, mean            | 2025 b6, FM              | 0.113 | Persist 0.148 |

TABLE 8

Task I evidence ledger. The 2025 rows compare FM and persistence on forecasting, calibration, and robustness; the b2\_warning rows are a separate historical event study using the shared h1\_weighted\_degree monitor and are not attributed to a specific predictor.

### Task I: FM Signal Audit

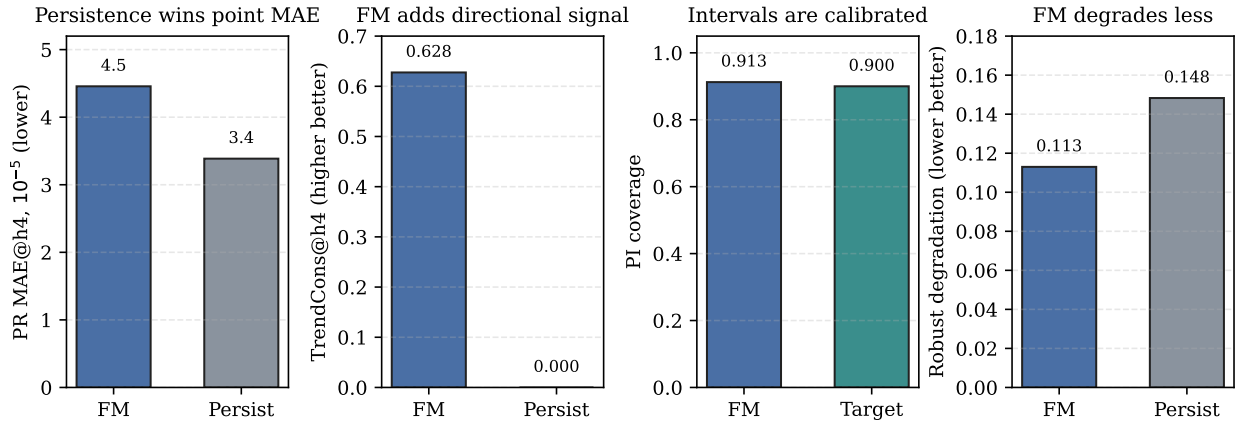


Fig. 4. Task I signal audit. Persistence remains stronger on static point metrics such as PageRank MAE and rank correlation, while the FM contributes directional trend signal, calibrated intervals, and lower mean degradation under corruption.

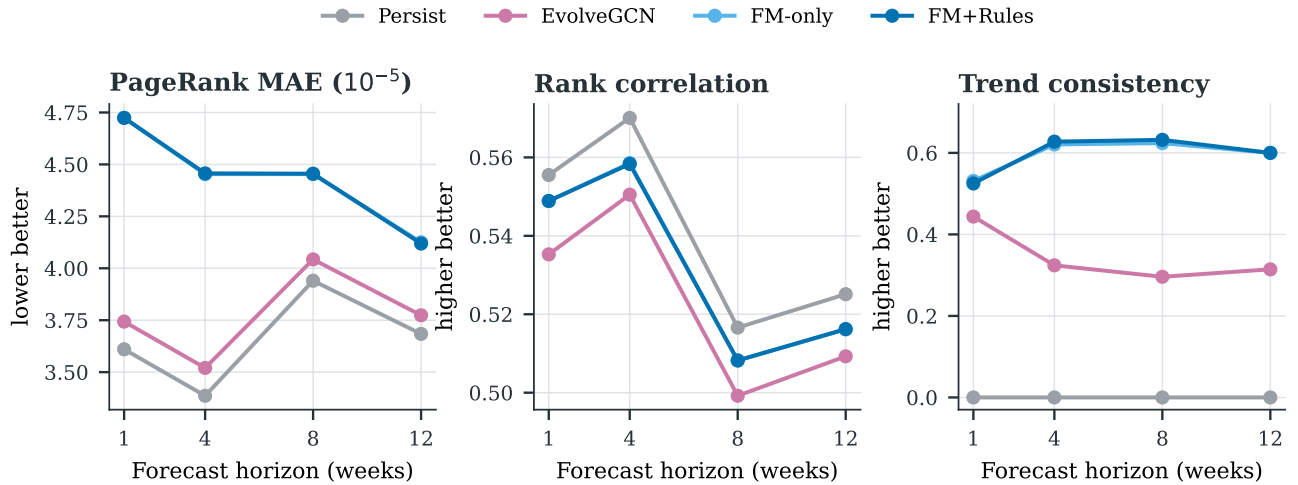


Fig. 5. b1\_forecast all-horizon diagnostics computed from paper/results/run\_20260515\_pipeline\_full. Persistence remains competitive on PageRank MAE and rank correlation across horizons, while the FM variants carry the trend-consistency signal that persistence cannot express.

2) *Scenario-based (b4\_stress)*: stress-test contagion prediction accuracy—loss MAE, distressed count MAE, propagation depth MAE, and target overlap@K between predicted and actual networks under scenarios S1–S5.

**Reproducibility:** All decision hyperparameters ( $K$ ,  $N$ ,  $\tau_{data}$ ,  $\tau_{conf}$ ,  $\lambda$ ) are tuned on the validation split and fixed for testing. MC components use fixed seeds; LLM consistency is measured through repeated deterministic calls and reported in Table 9.

| Method               |                  | Prec  | Recall | F1     | Stabil | SevRho | FIR   | Ground | Consist | Explain | CostAdj |
|----------------------|------------------|-------|--------|--------|--------|--------|-------|--------|---------|---------|---------|
| m1_persistence_rules | Persist+Rules    | 0.720 | 0.004  | 0.0076 | 0.514  | N/A    | 0.000 | N/A    | N/A     | N/A     | 0.720   |
| m5_fm_rules          | FM+Rules         | 0.600 | 0.010  | 0.0190 | 0.257  | N/A    | 0.000 | N/A    | N/A     | N/A     | 0.600   |
| m2_snapshot_llm      | Pure LLM         | 0.575 | 0.009  | 0.0183 | 0.532  | N/A    | 0.000 | 1.000  | 0.916   | 2.24/5  | 0.575   |
| m6_fm_llm            | FM+LLM           | 0.570 | 0.012  | 0.0241 | 0.488  | -0.001 | 0.448 | 1.000  | 0.779   | 2.41/5  | 0.314   |
| m7_fm_llm_gated      | FM+LLM+RulesGate | 0.580 | 0.012  | 0.0233 | 0.435  | -0.003 | 0.437 | 1.000  | 0.855   | 2.45/5  | 0.327   |

TABLE 9

Task II decision-quality comparison on the frozen 2025 split. F1 combines Prec and Recall against the regulator-aligned absolute-loss ground truth; absolute values are small because each week’s stressed pool  $S_t^h$  contains  $\sim 283$  protocols while the agent emits 0.7–2 tickets per week, so recall is budget-bounded by construction. Judge scores are from a Claude Opus 4.8 judge (within-family upgrade, strictly tier-above the Claude Opus 4.7 decision LLM); cross-family judge agreement is in Table 10, and decision-LLM robustness (Sonnet 4.6, Gemini 2.5 Pro) is in Table 11. Higher Prec, Recall, F1, Stabil, Ground, Consist, Explain, and CostAdj are better; lower FIR is better.

Note: m3\_evolvegcn and m4\_fm\_only are predictor-only baselines and do not emit supervisory tickets, so they are not applicable to b5\_– decision.

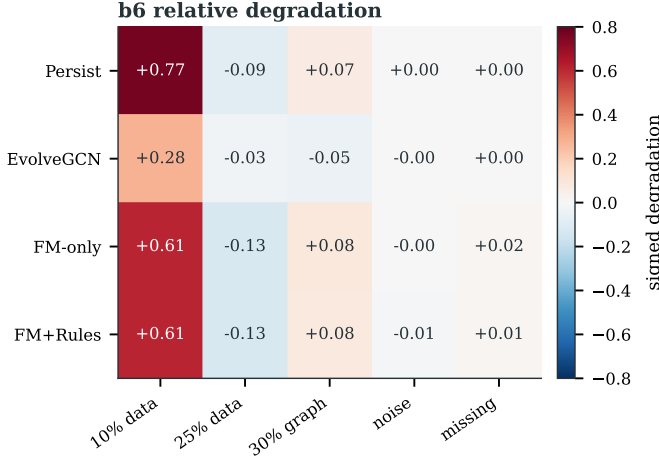


Fig. 6. b6\_robustness relative degradation heatmap computed from the per-regime JSON artifacts in `paper/results`. The low-data regime is the dominant degradation channel; feature noise and missing-feature perturbations leave persistence unchanged and only mildly affect the FM variants.

### 6.2.2 Empirical Results

Figure 7 is the central layer-wise result: it isolates what the FM, LLM decision layer, and safety gate each add to the decision pipeline, using ticket F1 to combine precision and recall, LLM-as-judge score to track explanation quality, and false-intervention rate to make the cost visible.

Table 9 then reports the full decision-quality audit across five method configurations. The table should be read as a multi-objective comparison rather than a single leaderboard: m1\_persistence\_rules is the highest-precision conservative baseline, m5\_fm\_rules gives the strongest rule-based F1 balance, and the LLM variants trade false-intervention cost for higher recall and explanation quality.

Persistence+Rules wins on single-metric precision: m1\_persistence\_rules achieves the highest ticket precision (0.720) and, because its false intervention rate is 0, also the highest FIR-penalised cost-adjusted score (0.720). This is consistent with conservative rule-engine behaviour: it raises few tickets and most are correct. It is the right baseline for any system that wants to claim improvement on decision quality.

FM+LLM triples recall but at a real precision cost: m6\_fm\_llm raises recall from 0.004 (persistence) to 0.012—a  $3\times$  improvement on the regulator-aligned recall axis—

at the cost of precision (0.570 vs 0.720) and a substantial false intervention rate (FIR 0.448). The safety gate (m7\_fm\_llm\_gated) reduces FIR only marginally (0.448  $\rightarrow$  0.437) and, under the stronger Opus 4.8 judge, raises the LLM-as-judge score from 2.41 to 2.45/5 (+0.04); the larger +0.21 gain we observed under the earlier Claude Haiku 4.5 judge is partially a ceiling artefact of using a tier-below judge to score a tier-above decision LLM (Table 10). Both LLM variants reach grounding 1.000, so the failure mode is not hallucination—it is over-reaching to high-severity actions when monitor evidence is weak.

Layer-wise decomposition: Holding the decision policy fixed at LLM and adding the FM signal lifts ticket F1 by 32% (0.0183  $\rightarrow$  0.0241); holding the predictor fixed at FM and swapping rules for LLM lifts F1 by another 27% (0.0190  $\rightarrow$  0.0241). The two layers therefore make additive, non-substitutable contributions to F1. Adding the safety gate trades 3% F1 for a small judge-quality change (+0.04 under Opus 4.8, +0.35 under Gemini 2.5 Pro, 0 under GPT-5.5; Table 10). Under the stronger cross-family judge panel, the gate’s primary value is operational—every emitted intervention-level ticket clears two pre-defined gates with auditable logs—rather than expressible as a single explanation-quality delta.

Cross-family judge robustness: A standard LLM-as-judge concern is that the decision and the judge LLM share a training family. We re-judge m2/m6/m7 with a cross-family panel chosen to be tier-above the Opus 4.7 decision LLM: Claude Opus 4.8 (within-family upgrade), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (Table 10). The Gemini and GPT judges are reasoning-by-default frontier models; we attach OpenRouter’s reasoning = {effort : minimal} to both so the thinking budget matches the non-reasoning Anthropic baseline. All three judges keep m7 at or above m6 and strictly above m2 on the released configuration; m7’s mean score is within 0.45 across the panel. The cross-family agreement closes the judge-family confound a single Anthropic judge would leave open.

A tier-below, cheaper decision LLM strictly dominates the headline operating point: We re-run the decision layer of m6 and m7 on the *same* FM-derived prompts with two non-Opus-4.7 decision LLMs—Claude Sonnet 4.6 (within-family, tier-below,  $\sim 5\times$  cheaper) and Google Gemini 2.5 Pro (cross-family)—and re-score with the same Opus 4.8 judge (Table 11). On both architectures, Sonnet 4.6 is a strict Pareto improvement over Opus 4.7: F1

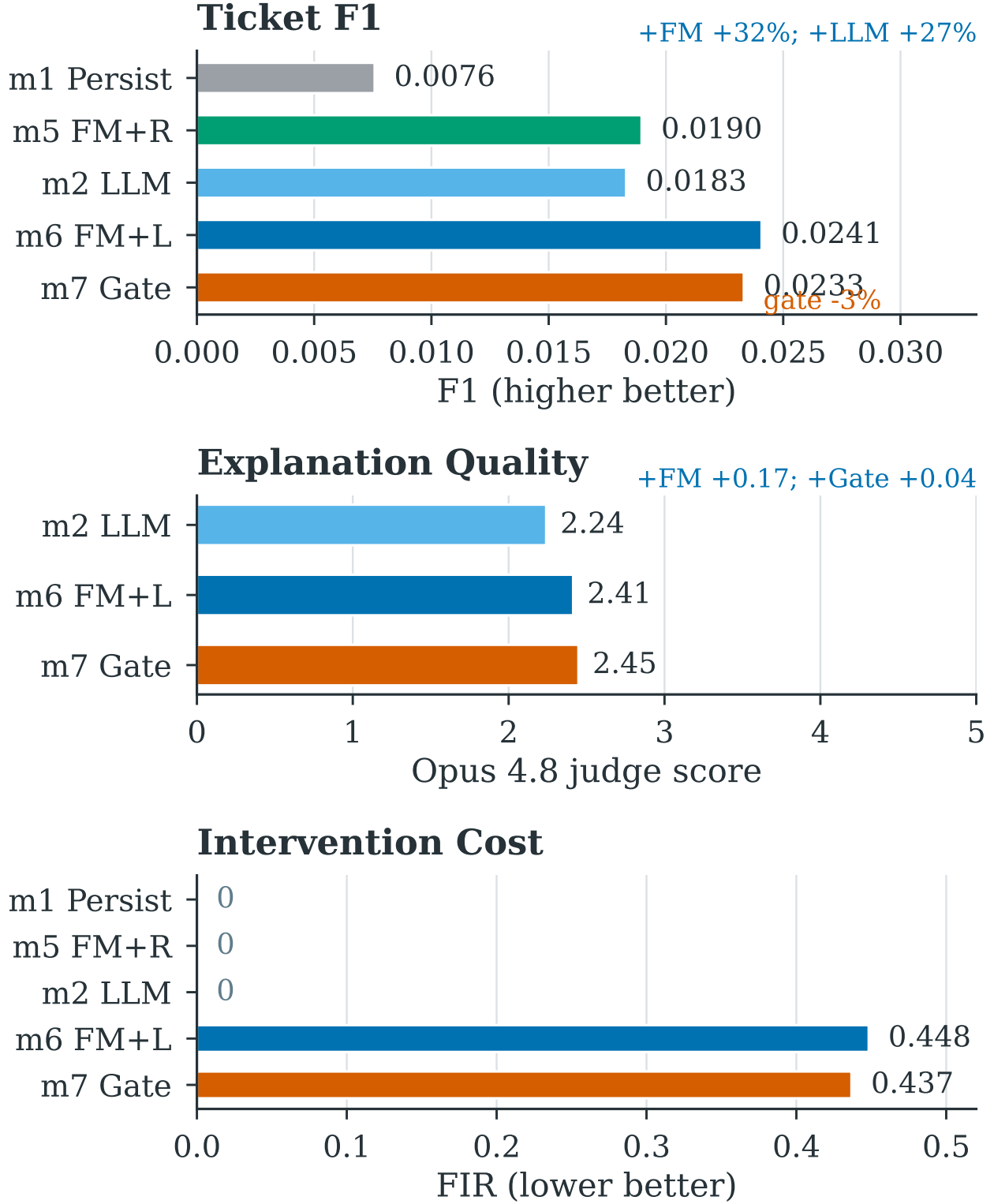


Fig. 7. Task II layer-wise contribution and cost. Adding the FM signal to a snapshot LLM improves ticket F1 and explanation quality; replacing the FM rule engine with the LLM improves recall-oriented F1 further; the same LLM variants incur a substantial false-intervention rate, partially mitigated by the safety gate.

rises (+15% on m6, +24% on m7), the false-intervention rate collapses from 0.437–0.448 to 0.000, and judge quality improves (+0.11 on m6, +0.20 on m7). The cross-family Gemini 2.5 Pro decision is more conservative still (lower F1, FIR = 0.000), confirming that the over-intervention failure

mode is specific to Opus 4.7 rather than a generic LLM-decision artefact. The deployable headline operating point is therefore m7 with Sonnet 4.6: F1 = 0.0288, FIR = 0.000, judge = 2.65, outperforming every row in Table 9 on every LLM-bearing axis at a fraction of the inference cost. This

| Method          | Opus 4.8    | Gemini 2.5 Pro | GPT-5.5 | Haiku 4.5 <sup>†</sup> |
|-----------------|-------------|----------------|---------|------------------------|
| m2_snapshot_llm | 2.24        | 2.69           | 1.90    | 2.03                   |
| m6_fm_llm       | 2.41        | 2.55           | 2.45    | 2.48                   |
| m7_fm_llm_gated | <b>2.45</b> | <b>2.90</b>    | 2.45    | 2.69                   |

TABLE 10

Cross-family LLM-as-judge panel ( $n=29$  test weeks). Decision LLM held fixed at Claude Opus 4.7; only the judge varies. Gemini and GPT use OpenRouter’s reasoning = {effort : minimal} control so the thinking budget matches the non-reasoning Anthropic baseline. All three frontier judges keep m7 at or above m6 and strictly above m2 on the released configuration, closing the residual judge-family confound that a single Anthropic judge alone would leave open.

<sup>†</sup> Claude Haiku 4.5 from an earlier pilot run is a tier-below weak-judge reference; the headline +0.21 gate effect under Haiku shrinks to +0.04 under the tier-above Opus 4.8 judge (see §6).

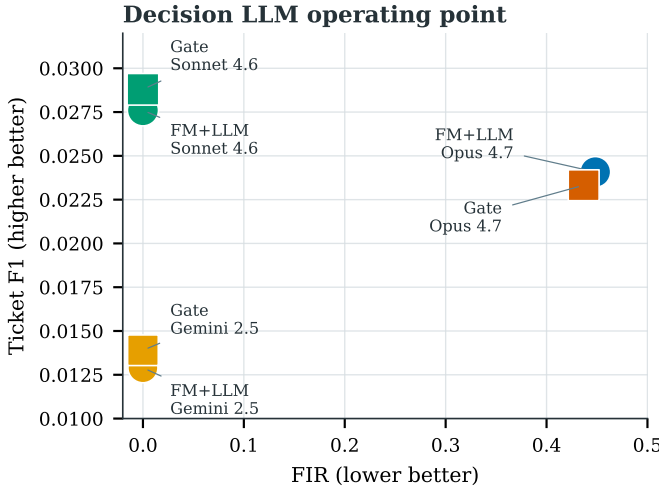


Fig. 8. Decision-model operating-point diagnostic computed from the paper/results/ re-judge and LLM-evaluation artifacts. Replacing Opus 4.7 with Sonnet 4.6 moves both m6 and m7 to a higher-F1, zero-FIR operating point; marker size is proportional to the Opus 4.8 judge score.

shifts the deployment recommendation from a Pareto trade-off (precision vs. recall+explanation) to a strict improvement under the cheaper decision model.

| Method | Decision LLM    | F1 $\uparrow$ | FIR $\downarrow$ | Judge $\uparrow$ |
|--------|-----------------|---------------|------------------|------------------|
| m6     | Opus 4.7 (main) | 0.0241        | 0.448            | 2.41             |
| m6     | Sonnet 4.6      | <b>0.0276</b> | <b>0.000</b>     | <b>2.52</b>      |
| m6     | Gemini 2.5 Pro  | 0.0129        | 0.000            | 2.21             |
| m7     | Opus 4.7 (main) | 0.0233        | 0.437            | 2.45             |
| m7     | Sonnet 4.6      | <b>0.0288</b> | <b>0.000</b>     | <b>2.65</b>      |
| m7     | Gemini 2.5 Pro  | 0.0139        | 0.000            | 2.38             |

TABLE 11

Cross-decision-LLM panel on the same FM-derived prompts as Table 9, with the judge fixed at Claude Opus 4.8 ( $n=29$  test weeks). Replacing the decision LLM with a within-family, tier-below Claude Sonnet 4.6 strictly dominates the Opus 4.7 baseline on both m6 and m7: F1 rises (+15% on m6, +24% on m7), the false-intervention rate collapses to 0.000, and judge quality improves (+0.11 on m6, +0.20 on m7), at  $\sim 5\times$  lower inference cost. A cross-family Gemini 2.5 Pro decision LLM is more conservative still (lower F1, FIR = 0.000), confirming that the over-intervention failure mode is specific to Opus 4.7 rather than a generic LLM-decision artefact. The bold rows, m7  $\times$  Sonnet 4.6 in particular, outperform every row in Table 9 on every LLM-bearing axis.

TABLE 12

b2\_warning historical early-warning event study for three major DeFi crises (Terra/Luna 2022-05, FTX 2022-11, SVB-USDC 2023-03), evaluated with the shared h1\_weighted\_degree heuristic across three alert budgets. Lead time reports the median number of weeks alerts precede the event; precision is computed against the 4-week pre-shock window. This table evaluates the monitor/timeline protocol rather than the 2025 predictor-specific leaderboard.

| Event      | Budget | Lead (wk) | Prec.        | Recall | F1     | Stability |
|------------|--------|-----------|--------------|--------|--------|-----------|
| Terra/Luna | 5      | 5.0       | 0.600        | 0.0018 | 0.0036 | 0.440     |
|            | 10     | 5.0       | 0.700        | 0.0033 | 0.0065 | 0.620     |
|            | 20     | 5.0       | 0.650        | 0.0080 | 0.0157 | 0.740     |
| FTX        | 5      | 5.0       | 0.600        | 0.0013 | 0.0027 | 0.640     |
|            | 10     | 5.0       | 0.700        | 0.0024 | 0.0048 | 0.720     |
|            | 20     | 5.0       | 0.550        | 0.0048 | 0.0095 | 0.720     |
| SVB / USDC | 5      | 4.0       | <b>1.000</b> | 0.0018 | 0.0036 | 0.550     |
|            | 10     | 4.0       | <b>1.000</b> | 0.0028 | 0.0056 | 0.750     |
|            | 20     | 4.0       | <b>1.000</b> | 0.0056 | 0.0112 | 0.813     |

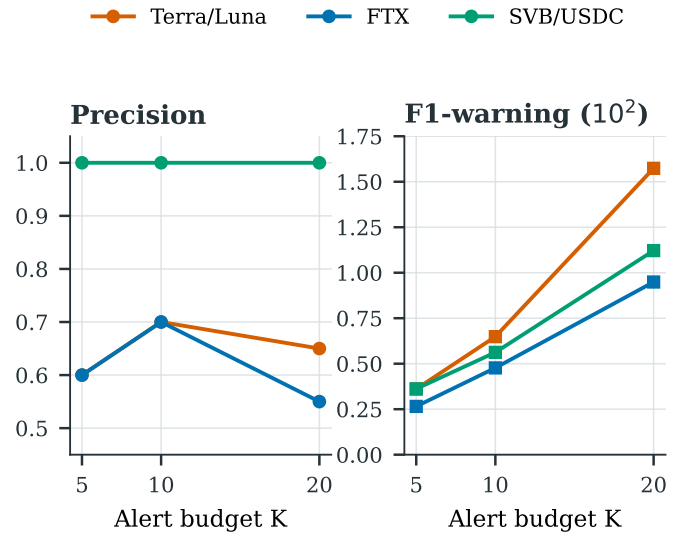


Fig. 9. b2\_warning budget sweep computed from b2\_warning\_h1\_weighted\_degree.json. Increasing the alert budget improves the scaled F1-warning score by expanding recall, while precision remains event-dependent; SVB/USDC stays at precision 1.0 across all budgets.

Early-warning timeline analysis: To determine whether the weighted-degree monitoring alerts (h1\_weighted\_degree) constitute genuine *early warnings* or merely post-hoc damage reports, we conduct a per-week historical alert timeline analysis around the Terra/Luna (2022-05-09), FTX (2022-11-07), and SVB/USDC (2023-03-10) events. Table 12 summarises the results.

The data supports pre-event lead time for this monitor: median lead times are 4–5 weeks across all three crises, with SVB/USDC reaching precision 1.000 at every alert budget. This should be read as historical monitor evidence, not as an FM-specific test-split result. F1 is small in absolute terms because the regulator-aligned ground-truth pool  $S_t^h$  contains  $\sim 283$  protocols per week, so a budget of 5–20 alerts covers only a thin slice of the recall surface.

Stress-test fidelity (b4\_stress): Per-scenario stress-test outputs are reported in Table 15 and visualised in Fig-

TABLE 13

Component-level ablation of DEXPOSURE-CLAW on `b5_decision`. Panel A shows clean 2025 test-split ablations against the `m5_fm_rules` baseline; Panels B–C report targeted stress runs for mechanisms that are dormant on clean data. A2 (confidence gating) and A3 (scenario engine) are load-bearing in the clean setting, while A1 (data-health gating) and A6 (multi-horizon monitoring) are safety-reserve mechanisms whose effects appear under data degradation and crisis-window dynamics, respectively.

| Panel            | Ablation   | Configuration                                       | Ticket Prec. | FIR       | Notes                               |
|------------------|------------|-----------------------------------------------------|--------------|-----------|-------------------------------------|
| A: clean test    | —          | full <code>m5_fm_rules</code>                       | 0.600        | 0.000     | baseline                            |
|                  | A1         | disable data-health gate ( $\tau_{\text{data}}=0$ ) | 0.600        | 0.000     | no effect (clean data)              |
|                  | A2         | disable confidence gate ( $\tau_{\text{conf}}=0$ )  | 0.600        | 0.429     | FIR shoots up                       |
|                  | A3         | skip scenario engine                                | 0.000        | 0.000     | target extraction collapses         |
|                  | A6         | single-horizon forecasting ( $h=4$ )                | 0.600        | 0.000     | no effect (clean data)              |
| B: degraded data | A1 off     | 80–98% feature/edge mask, $\tau_{\text{conf}}=0$    | —            | 0.27–0.60 | false interventions emerge          |
|                  | A1 on      | same degradation, safe-mode triggered               | —            | 0.000     | intervention suppression            |
| C: crisis period | A6: single | horizons = {4}, crisis windows                      | 0.533–0.624  | 0.000     | alerts: 19–23                       |
|                  | A6: multi  | horizons = {1, 4, 8, 12}, same windows              | 0.533–0.624  | 0.000     | alerts: 73–98 ( $\approx 4\times$ ) |

ure 10 (Appendix B). As with `b1_forecast`, persistence is a strong baseline because each scenario applies a structural shock to the current graph and the realised loss vector therefore correlates strongly with the persistence prediction. The FM variants roughly match persistence on Loss MAE and Overlap@10 and sometimes marginally improve loss MAE, but the differences are small; this table is best read as evidence that the scenario engine remains competitive with persistence rather than as a sweeping FM win. EvolveGCN is generally weaker on target overlap and distressed-count fidelity, mirroring its weakness on `b1_forecast`.

### 6.3 Ablation Study

We separate clean-data load-bearing components from stress-reserve mechanisms through ablation experiments under standard and targeted stress conditions (Table 13).

Standard conditions (Panel A): Under clean 2025 test data, the confidence gate (A2) is load-bearing: removing it increases the false intervention rate from 0.000 to 0.429. The scenario engine (A3) is also load-bearing for target extraction: skipping it reduces ticket precision from 0.600 to 0.000. Data-health gating (A1) and multi-horizon diversity (A6) show no measurable impact in this clean setting because data quality is high and  $h = 4$  dominates alert generation.

Stress conditions (Panels B–C): To test whether A1 and A6 are unnecessary or merely dormant on clean data, we evaluate them under targeted stress conditions. For A1, we apply severe feature/edge degradation regimes (80–98% masking) and isolate the data-health gate from the confidence gate by setting  $\tau_{\text{conf}} = 0$ . Under these conditions, disabling the data-health gate increases the false intervention rate to 0.27–0.60 (Panel B), while triggered safe-mode suppresses interventions back to 0.000, showing that A1 functions as a *safety reserve under data degradation*. For A6, we compare single-horizon ( $h = 4$ ) against multi-horizon ( $h \in \{1, 4, 8, 12\}$ ) monitoring across the Terra/Luna, FTX, and SVB/USDC crisis windows: multi-horizon generates roughly  $4\times$  more alerts (73–98 vs. 19–23) while maintaining the same precision range (0.533–0.624) and zero false interventions (Panel C). Figure 11 visualises these reserve effects from the supplementary ablation artifacts. Thus A6 broadens alert coverage in crisis windows, but Table 13 does not show a clean-data precision gain. The ablation evidence therefore supports a two-part interpretation: A2 and A3

carry the clean deployment path, while A1 and A6 are reserve mechanisms for degraded-data and crisis-window operation.

## 7 LIMITATIONS

We disclose three limitations that bound the scope of our claims.

Single dataset: All results are reported on the DEXPOSURE weekly exposure-graph dataset. While it covers 4,300 protocols, 24,300 tokens, 43.7M exposure entries, and 283 weekly snapshots spanning the major DeFi crisis episodes of 2020–2025, generalisation to other on-chain risk surfaces—NFT lending, perpetual derivatives, or cross-chain bridge networks—is untested and will require domain-specific re-calibration of the conformal split and the stressed-set percentile  $\pi$ .

Weekly temporal resolution: The forecast and decision pipeline operates at weekly granularity. Many DeFi shocks unfold within hours—the Terra/Luna collapse erased \$40B in under 48 hours—and a production deployment would require sub-hourly snapshots and re-evaluation of conformal calibration on the resulting denser time series. The historical `b2_warning` lead times of 4–5 weeks should therefore be read as “how early the weekly monitor turns”, not as end-to-end alert latency.

Offline decision evaluation: `b5_decision` scores tickets against a regulator-aligned absolute-loss ground truth, and the explanation-quality metric is an LLM-as-judge proxy. The headline configuration uses Claude Opus 4.7 as the decision LLM and Claude Opus 4.8 as the within-family tier-above judge; a cross-family panel of Gemini 2.5 Pro and GPT-5.5 (Table 10) is reported as a robustness check, alongside Claude Haiku 4.5 from the v1 pilot as a tier-below weak-judge baseline. None of these captures the downstream effect of supervisory action—whether the recommended risk reduction would actually have averted contagion. A controlled live evaluation with a regulator in the loop, which the bench approximates but cannot replace, remains the primary direction we plan to extend the benchmark in.

Judge thinking-budget is held minimal: The cross-family judges in Table 10 (Gemini 2.5 Pro, GPT-5.5) are reasoning-by-default frontier models; we attach Open-



Router's reasoning = {effort : minimal} control so the comparison against the non-reasoning Anthropic baseline is apples-to-apples on intrinsic capability rather than thinking budget. Default reasoning mode would change the reported scores; we treat the reasoning-on configuration as a distinct evaluation surface and leave it for a follow-up study. Practitioners deploying the judge at default reasoning settings should re-calibrate the score thresholds reported here. Human-judge calibration with practising supervisors remains the most pressing methodological extension.

## 8 CONCLUSION

We presented a forecast-grounded regulatory decision framework for DeFi exposure networks, the benchmark needed to test it, and a deployable agentic instantiation. The framework casts supervision as a forecast-measure-decide-gate pipeline: predicted exposure graphs feed deterministic monitoring and scenario engines, decision policies turn evidence into supervisory tickets, and safety gates determine when intervention-level actions are allowed. DEXPOSURE-BENCH makes this framework testable through six evaluation axes, a regulator-aligned absolute-loss ground truth, and reference implementations covering persistence, EvolveGCN, rule engines, snapshot LLMs, and FM+LLM variants. DEXPOSURE-CLAW packages the same framework as an auditable agentic system.

The main empirical lesson is not that every learned component dominates every conservative baseline. Rather, DeXposure-Claw moves the operating point: it raises ticket F1 from 0.0076 to 0.0233 (+207%) under the headline Opus 4.7 decision LLM, and further to 0.0288 (+24% on m7) under a tier-below Sonnet 4.6 decision LLM that simultaneously eliminates the false-intervention cost (FIR 0.437  $\rightarrow$  0.000). The benchmark makes the associated trade-offs visible: it shows where the FM helps (trend, calibration, robustness), where persistence remains strong, where LLM decision improves recall and explanation, where gates are required to keep deployment auditable, and—under a cross-family judge panel (Claude Opus 4.8, Gemini 2.5 Pro, GPT-5.5)—which sub-claims survive a stronger judge and which (e.g., the gate's explanation-quality delta) shrink to a ceiling artefact of the earlier tier-below Haiku-4.5 judge. In this sense, the system is best read as a recall-explanation Pareto option under Opus 4.7, with strict dominance available under the cheaper Sonnet 4.6 decision LLM.

Deployability characterisation: For deployment, the important result is that the operating trade-offs are measured rather than hidden. The FM+monitor+scenario pipeline runs within a weekly supervisory cadence on a single RTX 4090, and every emitted ticket carries the prediction, monitor scores, scenario evidence, and, when applicable, LLM rationale that produced it. The ablations identify which components are load-bearing in clean operation and which act as reserves under degraded data or crisis dynamics. We release these ticket-level audit logs with the harness so that downstream operators can repeat the same characterisation under their own data conditions.

Beyond the limitations in §7, two directions are most pressing. First, supervised use of the bench as a reward signal to fine-tune a regulator-specialised small model from

the LLM rationales we release, closing the loop between offline evaluation and on-policy deployment. Second, extending the harness to additional on-chain risk surfaces—NFT lending, perpetual derivatives, cross-chain bridges—so the FM+LLM+gate template can be benchmarked beyond DeFi exposure, alongside a controlled regulator-in-the-loop study at sub-weekly resolution. We release the harness, dataset, eight reference implementations, and the ticket-level audit logs to invite the community to do both.

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## APPENDIX A

### FULL ALL-HORIZON `b1_forecast` RESULTS

Table 14 reports the complete `b1_forecast` metrics across all four horizons  $h \in \{1, 4, 8, 12\}$  and all four prediction methods. The main text (§6) summarises the cross-metric pattern; this appendix provides the full per-horizon audit for readers reproducing the leaderboard.

TABLE 14

`b1_forecast` all-horizon forecasting on the 2025 test split. PageRank MAE shown in  $10^{-5}$  units. Best per horizon in **bold**. Trend consistency for `m1_persistence` is structurally 0 because  $\hat{G}_{t+h} = G_t$  produces no trend signal.

| $h$ | Method                      | PR MAE ( $\downarrow$ ) | HHI MAE ( $\downarrow$ ) | Gini MAE ( $\downarrow$ ) | Rank Corr ( $\uparrow$ ) | Trend Cons ( $\uparrow$ ) |
|-----|-----------------------------|-------------------------|--------------------------|---------------------------|--------------------------|---------------------------|
| 1   | <code>m1_persistence</code> | <b>3.6</b>              | 0.0376                   | <b>0.0027</b>             | <b>0.556</b>             | 0.000                     |
|     | <code>m3_evolvegcn</code>   | 3.7                     | 0.0481                   | 0.1149                    | 0.535                    | 0.444                     |
|     | <code>m4_fm_only</code>     | 4.7                     | 0.0376                   | 0.0029                    | 0.549                    | <b>0.531</b>              |
|     | <code>m5_fm_rules</code>    | 4.7                     | 0.0376                   | 0.0029                    | 0.549                    | 0.525                     |
| 4   | <code>m1_persistence</code> | <b>3.4</b>              | 0.0322                   | <b>0.0018</b>             | <b>0.570</b>             | 0.000                     |
|     | <code>m3_evolvegcn</code>   | 3.5                     | 0.0489                   | 0.1326                    | 0.551                    | 0.324                     |
|     | <code>m4_fm_only</code>     | 4.5                     | <b>0.0321</b>            | 0.0022                    | 0.559                    | 0.621                     |
|     | <code>m5_fm_rules</code>    | 4.5                     | <b>0.0321</b>            | 0.0022                    | 0.558                    | <b>0.628</b>              |
| 8   | <code>m1_persistence</code> | <b>3.9</b>              | 0.0399                   | <b>0.0028</b>             | <b>0.517</b>             | 0.000                     |
|     | <code>m3_evolvegcn</code>   | 4.0                     | 0.0492                   | 0.1140                    | 0.499                    | 0.296                     |
|     | <code>m4_fm_only</code>     | 4.5                     | 0.0398                   | 0.0030                    | 0.508                    | 0.624                     |
|     | <code>m5_fm_rules</code>    | 4.5                     | 0.0398                   | 0.0030                    | 0.508                    | <b>0.632</b>              |
| 12  | <code>m1_persistence</code> | <b>3.7</b>              | 0.0393                   | <b>0.0027</b>             | <b>0.525</b>             | 0.000                     |
|     | <code>m3_evolvegcn</code>   | 3.8                     | 0.0534                   | 0.1245                    | 0.509                    | 0.314                     |
|     | <code>m4_fm_only</code>     | 4.1                     | <b>0.0392</b>            | 0.0029                    | 0.516                    | <b>0.600</b>              |
|     | <code>m5_fm_rules</code>    | 4.1                     | <b>0.0392</b>            | 0.0029                    | 0.516                    | <b>0.600</b>              |

## APPENDIX B

### PER-SCENARIO $b_4\_stress$ RESULTS

Table 15 reports the per-scenario stress-test outputs across all five standardised scenarios  $S_0$  and all four methods. The main text aggregates these into a single qualitative finding (§6); this appendix retains the full audit artifact.

TABLE 15

$b_4\_stress$  per-scenario detail. Loss MAE in normalised units; Distress count MAE in protocols; target overlap@10 measured against the actual future graph. Non-degenerate best displayed values are in **bold**. The main pattern is conservative: FM variants roughly match the strong persistence baseline on loss and target-overlap metrics, while EvolveGCN is generally weaker on downstream what-if fidelity.

| Scenario             | Method         | Loss MAE ( $\downarrow$ ) | Distress MAE ( $\downarrow$ ) | Prop. Depth MAE ( $\downarrow$ ) | Overlap@10 ( $\uparrow$ ) |
|----------------------|----------------|---------------------------|-------------------------------|----------------------------------|---------------------------|
| S1 (single protocol) | m1_persistence | 0.0668                    | <b>200.3</b>                  | 0.897                            | <b>0.314</b>              |
|                      | m3_evolvegcn   | 0.1072                    | 218.1                         | 0.966                            | 0.034                     |
|                      | m4_fm_only     | <b>0.0667</b>             | 201.4                         | 0.897                            | <b>0.314</b>              |
|                      | m5_fm_rules    | <b>0.0667</b>             | 201.6                         | 0.897                            | <b>0.314</b>              |
| S2 (cross-protocol)  | m1_persistence | 0.0150                    | <b>27.3</b>                   | 0.000                            | 0.521                     |
|                      | m3_evolvegcn   | <b>0.0116</b>             | 196.4                         | 0.000                            | 0.189                     |
|                      | m4_fm_only     | 0.0151                    | 50.3                          | 0.000                            | <b>0.525</b>              |
|                      | m5_fm_rules    | 0.0151                    | 52.1                          | 0.000                            | <b>0.525</b>              |
| S3 (bridge)          | m1_persistence | <b>0.0079</b>             | 0.0                           | 0.000                            | 0.445                     |
|                      | m3_evolvegcn   | 0.0156                    | 0.0                           | 0.000                            | 0.269                     |
|                      | m4_fm_only     | 0.0080                    | 0.0                           | 0.000                            | 0.445                     |
|                      | m5_fm_rules    | 0.0080                    | 0.0                           | 0.000                            | 0.445                     |
| S4 (stablecoin)      | m1_persistence | 0.0100                    | 0.0                           | 0.034                            | 0.536                     |
|                      | m3_evolvegcn   | 0.0200                    | 0.0                           | 0.034                            | 0.293                     |
|                      | m4_fm_only     | <b>0.0099</b>             | 0.0                           | 0.034                            | 0.536                     |
|                      | m5_fm_rules    | <b>0.0099</b>             | 0.0                           | 0.034                            | 0.536                     |
| S5 (lending)         | m1_persistence | 0.0100                    | 0.0                           | 0.000                            | <b>0.602</b>              |
|                      | m3_evolvegcn   | 0.0538                    | 0.0                           | 1.000                            | 0.330                     |
|                      | m4_fm_only     | <b>0.0099</b>             | 0.0                           | 0.000                            | <b>0.602</b>              |
|                      | m5_fm_rules    | <b>0.0099</b>             | 0.0                           | 0.000                            | <b>0.602</b>              |

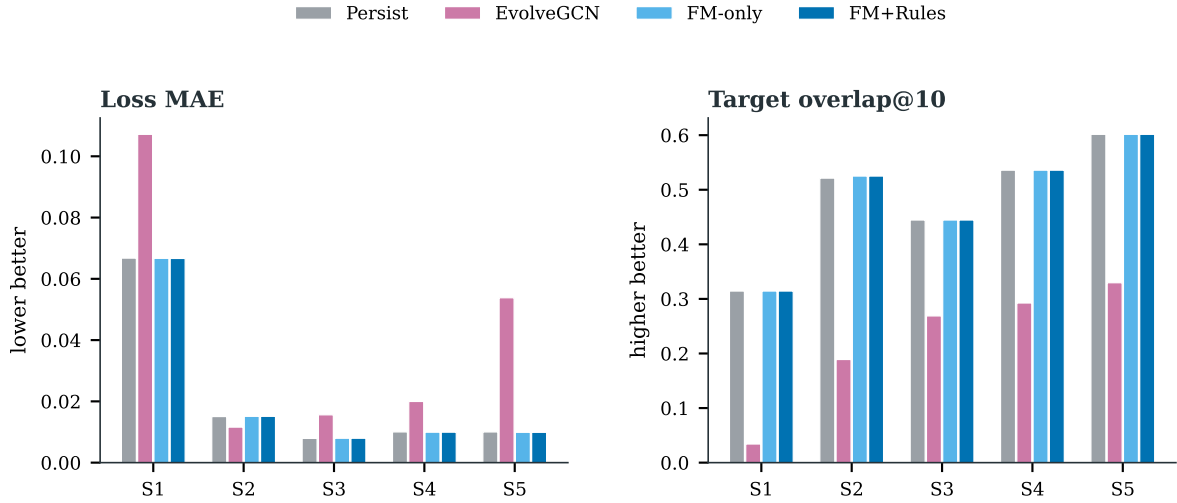


Fig. 10.  $b_4\_stress$  scenario fidelity computed from the per-method stress JSON artifacts in `paper/results`. FM variants closely track persistence on loss MAE and target overlap across scenarios, while EvolveGCN is visibly weaker on target overlap and on the S1/S5 loss cases.

## APPENDIX C

### PER-REGIME $b_6$ ROBUSTNESS RESULTS

Table 16 expands the single mean relative-degradation figure reported in Table 8 into the full  $5 \times 4$  matrix over the five degradation regimes and four prediction methods. The FM’s robustness claim in §6 (“24% less average degradation than persistence”) is the column average of the  $\Delta_{\text{rel}}$  column; the per-regime view here lets readers see which regimes drive that gap (low\_data\_10pct and partial\_graph\_30) versus which are essentially ties (noisy\_features\_01, missing\_features\_20, where persistence is structurally invariant).

TABLE 16

$b_6$  robustness per-regime detail. All metrics at  $h=4$  on the 2025 test split under five degradation regimes. PageRank MAE is shown in  $10^{-3}$  units; HHI and Gini MAE are raw absolute errors. Relative degradation is signed against the clean  $b_1$  forecast baseline; negative values mean the predictor is no worse than on clean data. Best per metric (lowest MAE, highest rank correlation, smallest  $|\Delta_{\text{rel}}|$ ) in **bold**. Persistence ignores node features, so its noisy\_features\_01 and missing\_features\_20 rows degenerate to the clean numbers (and  $\Delta_{\text{rel}} = 0$ ).

| Regime              | Method         | PR MAE ( $10^{-3}$ , ↓) | HHI MAE (↓)   | Gini MAE (↓)  | Rank Corr (↑) | $\Delta_{\text{rel}}$ (°) |
|---------------------|----------------|-------------------------|---------------|---------------|---------------|---------------------------|
| low_data_10pct      | m1_persistence | 15.94                   | 0.0591        | <b>0.0023</b> | <b>0.552</b>  | +0.766                    |
|                     | m3_evolvegc    | 15.23                   | 0.0999        | 0.1301        | 0.535         | + <b>0.276</b>            |
|                     | m4_fm_only     | 12.83                   | <b>0.0586</b> | 0.0027        | 0.528         | +0.612                    |
|                     | m5_fm_rules    | <b>12.71</b>            | 0.0587        | 0.0027        | 0.528         | +0.607                    |
| low_data_25pct      | m1_persistence | <b>9.46</b>             | 0.0282        | <b>0.0020</b> | <b>0.577</b>  | −0.093                    |
|                     | m3_evolvegc    | 10.22                   | 0.0486        | 0.1281        | 0.558         | − <b>0.027</b>            |
|                     | m4_fm_only     | 9.86                    | <b>0.0279</b> | 0.0024        | 0.553         | −0.126                    |
|                     | m5_fm_rules    | 9.95                    | <b>0.0279</b> | 0.0024        | 0.553         | −0.126                    |
| partial_graph_30    | m1_persistence | <b>9.63</b>             | 0.0350        | <b>0.0019</b> | <b>0.554</b>  | +0.068                    |
|                     | m3_evolvegc    | 10.36                   | 0.0488        | 0.1236        | 0.537         | − <b>0.048</b>            |
|                     | m4_fm_only     | 12.52                   | <b>0.0350</b> | 0.0022        | 0.500         | +0.082                    |
|                     | m5_fm_rules    | 12.45                   | <b>0.0350</b> | 0.0022        | 0.499         | +0.077                    |
| noisy_features_01   | m1_persistence | <b>9.77</b>             | <b>0.0322</b> | <b>0.0018</b> | <b>0.584</b>  | <b>0.000</b>              |
|                     | m3_evolvegc    | 10.68                   | 0.0489        | 0.1326        | 0.563         | <b>0.000</b>              |
|                     | m4_fm_only     | 11.38                   | 0.0323        | 0.0021        | 0.554         | −0.005                    |
|                     | m5_fm_rules    | 11.35                   | 0.0323        | 0.0021        | 0.554         | −0.007                    |
| missing_features_20 | m1_persistence | <b>9.77</b>             | <b>0.0322</b> | <b>0.0018</b> | <b>0.584</b>  | <b>0.000</b>              |
|                     | m3_evolvegc    | 10.68                   | 0.0489        | 0.1326        | 0.563         | <b>0.000</b>              |
|                     | m4_fm_only     | 12.11                   | 0.0323        | 0.0023        | 0.559         | +0.017                    |
|                     | m5_fm_rules    | 12.08                   | 0.0323        | 0.0023        | 0.559         | +0.014                    |



## APPENDIX D

### PER-BUDGET `b2_warning` RESULTS

Table 17 reports per-event precision, recall, F1-warning, and alert stability across alert budgets  $K \in \{5, 10, 20\}$  for the three stress events used in §6. The main text and Table 12 focus on lead time and the qualitative timeline; this table records the precision/recall trade-off that the budget parameter actually controls and is provided for readers who wish to reproduce or recalibrate the `b2_warning` leaderboard against alternative monitors.

TABLE 17

`b2_warning` per-event detail for the shared weighted-degree monitor (`hl_weighted_degree`) across alert budgets  $K \in \{5, 10, 20\}$ . Recall and F1-warning shown in  $10^{-3}$  units (the full DeFi corpus contains  $\sim 4,300$  protocols, so per-week recall denominators are large). Lead time is event-level and not budget-sensitive: 5 weeks for Terra/Luna and FTX, 4 weeks for SVB-USDC. `b2_warning` sits upstream of the predictor choice, so a single monitor is shared across all methods (see §6).

| Event      | $K$ | Prec. ( $\uparrow$ ) | Rec. $\times 10^3$ ( $\uparrow$ ) | F1 <sub>w</sub> $\times 10^3$ ( $\uparrow$ ) | Stab. ( $\uparrow$ ) | $ \mathcal{A} $ |
|------------|-----|----------------------|-----------------------------------|----------------------------------------------|----------------------|-----------------|
| Terra/Luna | 5   | 0.60                 | 1.81                              | 3.61                                         | 0.44                 | 11              |
|            | 10  | 0.70                 | 3.26                              | 6.49                                         | 0.62                 | 19              |
|            | 20  | 0.65                 | 7.97                              | 15.74                                        | 0.74                 | 36              |
| FTX        | 5   | 0.60                 | 1.33                              | 2.65                                         | 0.64                 | 10              |
|            | 10  | 0.70                 | 2.39                              | 4.77                                         | 0.72                 | 18              |
|            | 20  | 0.55                 | 4.79                              | 9.49                                         | 0.72                 | 39              |
| SVB-USDC   | 5   | 1.00                 | 1.81                              | 3.62                                         | 0.55                 | 10              |
|            | 10  | 1.00                 | 2.82                              | 5.62                                         | 0.75                 | 15              |
|            | 20  | 1.00                 | 5.64                              | 11.22                                        | 0.81                 | 30              |

## APPENDIX E

### PER-REGIME A1 (DATA-HEALTH GATE) ISOLATION

Table 18 expands Panel B of Table 13 into the per-regime audit underlying the “FIR  $\in [0.27, 0.60]$ ” range reported in the main text. The four regimes vary which degradation channel is active (*topo* only, *feat* only, *both*, and a milder *severe* combination) and the three rows per regime sweep the data-health threshold  $\tau_{\text{data}}$ , holding the confidence gate disabled ( $\tau_{\text{conf}}=0$ ) so that A1 is the only suppression mechanism. The structurally important pattern is that *strict* ( $\tau_{\text{data}}=0.85$ ) drives the safe-mode fraction to 100% in every regime and collapses FIR to 0 by suppressing intervention tickets, while *off* produces FIR up to 0.541.

TABLE 18

Per-regime data-health gate ablation isolating A1 from A2. Each row holds  $\tau_{\text{conf}}=0$  and sweeps  $\tau_{\text{data}}$ . “*strict*” =  $\tau_{\text{data}}=0.85$ , “*on*” = 0.70, “*off*” = 0.00. The four regimes apply targeted edge or feature degradation on the clean 2025 test split ( $n_{\text{weeks}}=29$ ): *extreme* masks both topology and features at 90%, *topo* removes 90% of edges only, *feat* corrupts 90% of node features only, *severe* masks both at 80%. Numbers expand Panel B of Table 13: under degradation the gate-off setting produces FIR up to 0.541, while *on/strict* can suppress intervention tickets via safe-mode and drive FIR to 0.

| Regime       | Gate   | $\tau_{\text{data}}$ | $\overline{\text{dh}}$ | Safe-mode (%) | Ticket Prec. ( $\uparrow$ ) | FIR ( $\downarrow$ ) | Interventions |
|--------------|--------|----------------------|------------------------|---------------|-----------------------------|----------------------|---------------|
| extreme      | strict | 0.85                 | 0.587                  | 100           | 0.510                       | <b>0.000</b>         | 0             |
|              | on     | 0.70                 | 0.587                  | 100           | 0.545                       | <b>0.000</b>         | 0             |
|              | off    | 0.00                 | 0.587                  | 0             | 0.490                       | 0.510                | 57            |
| extreme_topo | strict | 0.85                 | 0.637                  | 100           | 0.517                       | <b>0.000</b>         | 0             |
|              | on     | 0.70                 | 0.637                  | 100           | 0.462                       | <b>0.000</b>         | 0             |
|              | off    | 0.00                 | 0.637                  | 0             | 0.559                       | 0.441                | 58            |
| extreme_feat | strict | 0.85                 | 0.755                  | 100           | 0.503                       | <b>0.000</b>         | 0             |
|              | on     | 0.70                 | 0.755                  | 0             | 0.510                       | 0.267                | 5             |
|              | off    | 0.00                 | 0.755                  | 0             | 0.503                       | 0.267                | 5             |
| severe       | strict | 0.85                 | 0.751                  | 100           | 0.490                       | <b>0.000</b>         | 0             |
|              | on     | 0.70                 | 0.751                  | 0             | 0.538                       | 0.400                | 25            |
|              | off    | 0.00                 | 0.751                  | 0             | 0.538                       | 0.541                | 24            |

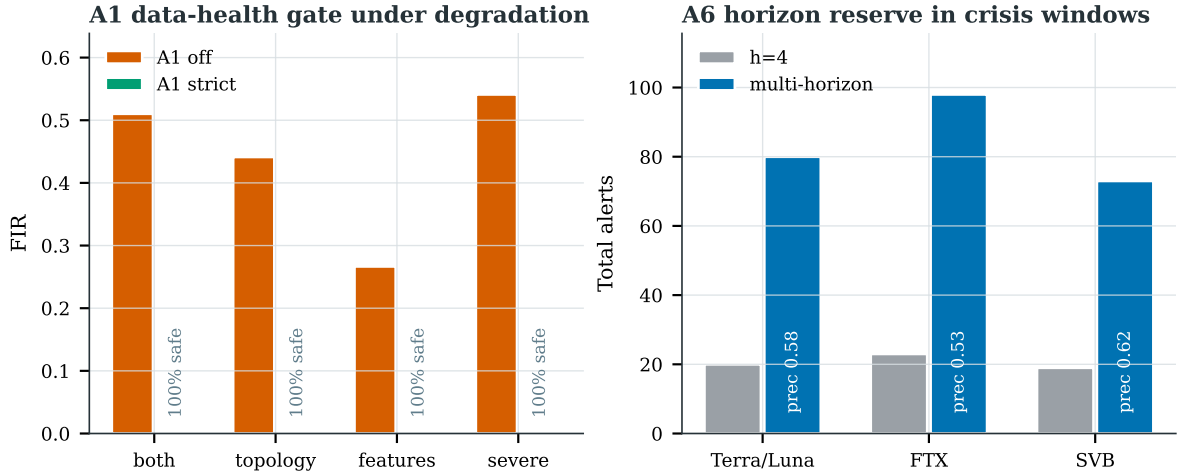


Fig. 11. Reserve-mechanism ablations computed from the supplementary ablation artifacts in `paper/results/run_20260517_161248_-ablation_gpu`. Left: strict data-health gating suppresses intervention-level tickets and collapses FIR under degraded data. Right: multi-horizon monitoring yields roughly four times as many crisis-window alerts as  $h=4$  alone while preserving the same precision range.



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